scientific reports

Check for updates

OPEN

Real-time deforestation anomaly detection using YOLO and LangChain agents for sustainable environmental monitoring

Shakti Kundu[1], Shalini Zanzote Ninoria[2], Ravi Prakash Chaturvedi[3], Annu Mishra[4], Akshat Agrawal[5], Reenu Batra[6], Mitiku Dubale[7]✉ & Arshad Hashmi[8]

Deforestation continues to pose a major threat to global ecosystems, biodiversity, and climate resilience, demanding intelligent and timely monitoring solutions. This study introduces a novel framework that integrates YOLOv8 (You Only Look Once) object detection with LangChain-based Agentic AI for real-time deforestation anomaly detection. The proposed system leverages YOLOv8's rapid and accurate visual recognition of deforestation indicators—such as tree stumps, logging machinery, and unauthorized human presence—while enhancing contextual reasoning and decision-making through LangChain agents. Extensive experiments using annotated satellite and drone imagery demonstrate steady improvements in training performance, with box_loss, cls_loss, and distribution focal loss reduced by more than 50%. Despite modest mean Average Precision (mAP50 ≈ 0.07), the integration of LangChain agents enabled dynamic threshold adjustment, reinforcement-learning-based feedback, and GIS-driven reporting, thereby reducing false positives and increasing recall (up to 24%) compared to baseline YOLO models. The framework not only provides actionable, geolocated alerts but also supports adaptive learning for evolving deforestation patterns. By combining the speed of deep learning with the autonomy of agentic AI, this work highlights a scalable, interpretable, and real-time approach for environmental monitoring. The findings establish a foundation for future research in multi-modal data fusion, edge deployment on drones and satellites, and sustainable forest management.

Keywords LangChain agent, YOLOv8 model, Anomaly detection, Deforestation monitoring, Environmental sustainability, Precision-Recall analysis

A major environmental problem, deforestation causes disturbance of ecosystems, climate change, and loss of species. Absorbing carbon dioxide, preserving water cycles, and giving many species homes—forests are vital. But mostly driven by agricultural development, urbanisation, and illicit logging, large-scale deforestation is fast increasing. Reports from environmental groups and the United Nations indicate that millions of hectares of forests are destroyed yearly, with dire effects on the ecology as well as the economy. Ground surveys and manual satellite image processing are two conventional approaches for tracking deforestation that are labour-intensive, time-consuming, and often ineffective in offering real-time observations[1]. By the time current deforestation detection methods identify signs of activity, irreversible environmental damage has already occurred. Furthermore difficulty to reach distant forest areas makes it tough for legislators and environmentalists to carry out timely measures[2].

[1]Department of Computer Science and Engineering, NIIT University, Neemrana 301705, Rajasthan, India. [2]College of Computing Sciences & IT, Teerthanker Mahaveer University, Moradabad, UP, India. [3]Department of Computer Science and Applications, Sharda University Greater Noida, Greater Noida, Uttar Pradesh, India. [4]Department of computer science and engineering, Sharda University Greater Noida, Greater Noida, Uttar Pradesh, India. [5]Department of Computer Science and Engineering, IILM University, Gurugram, Haryana, India. [6]School of Engineering and Technology, K. R. Mangalam University, Gurugram, Haryana, India. [7]Department of Public Health, College of Natural and Computational Sciences, Gambella University, Gambella, Ethiopia. [8]Department of Information Systems, Faculty of Computing and Information Technology in Rabigh (FCITR), King Abdulaziz University, Jeddah 21911, Saudi Arabia. ✉email: mitikudubalea@gmu.edu.et

Rising as a transforming instrument in environmental research, artificial intelligence (AI) provides creative ideas for resource protection and monitoring. Particularly those focused on image recognition, deep learning models have shown amazing ability in processing enormous volumes of satellite and aerial images. These algorithms can categorise changes in land cover, automatically find trends, and offer almost rapid updates on the state of the forests. With its real-time performance and great accuracy among several artificial intelligence-based object identification models, YOLOv8 (You Only Look Once) has attracted a lot of interest. YOLOV8v8, a convolutional neural network (CNN)-based model with fast object recognition and classification capability, is quite successful for the aim of deforestation detection. Unlike standard image-processing techniques, YOLOv8 processes a complete image in one pass, therefore drastically reducing computation time while keeping accuracy. Monitoring deforestation using YOLOv8 teaches the model on annotated satellite and aerial pictures wherein patches of deforested land are recognized as unique objects. Once taught, the model can automatically scan new images and highlight areas of deforestation. This capacity allows authorities to constantly monitor illegal activities such slash-and-burn farming, land encroachment, and logging from forests[3–6].

Artificial intelligence driven deforestation identification offers various advantages over more traditional monitoring methods. By permitting little human participation in large-scale research, it assures more accuracy and efficiency. Artificial intelligence models also interact with geospatial technologies like Geographic Information Systems (GIS) to demonstrate trends of deforestation over time. By use of this data-driven approach, governments and environmental groups develop targeted conservation strategies and carry out legislation intended to slow down forest loss. Deforestation detection driven by artificial intelligence poses challenges even with its advantages. Although high-resolution satellite photos might be expensive, cloud cover usually restricts woodland view, therefore lowering the utility of optical imaging models. Moreover, developing artificial intelligence models requires computer resources and large datasets—which would not be accessible in developing regions with highest rates of deforestation. Dealing with these challenges requires the use of synthetic aperture radar (SAR) imaging, another technology, to complement artificial intelligence-based detection[7].

Although artificial intelligence presents a good approach to preserve forests, ethical questions have to be answered to ensure responsible deployment. Data privacy, surveillance concerns, and artificial intelligence training model biases are under close supervision. Moreover, artificial intelligence models reflect a lot of processing capability, which raises questions about their carbon footprint. By means of model efficiency optimization and training deep learning model utilization of renewable energy, sustainable artificial intelligence approaches can help to lessen environmental consequences[8–10]. Many corporations and research institutes have successfully used artificial intelligence-based deforestation monitoring. For example, Google Earth Engine looks at satellite images using machine learning for variations in forest cover. Similarly, groups like the World Resources Institute use AI-powered systems like Global Forest Watch to provide real-time alerts of deforestation. These case studies demonstrate how effectively AI-driven monitoring systems identify and slow down world forest loss[11].

Emphasizing its accuracy, efficiency, and usefulness in practical settings, this paper aims to explore YOLOv8's prospects in the detection of deforestation. Extensive environmental monitoring driven by artificial intelligence will enable us to influence policy decisions, enhance conservation efforts, and assist sustainable land use practices[12]. The key focuses of further research should include enhancing model robustness, using multi-modal data sources, and developing reasonably priced artificial intelligence solutions for world forest preservation.

There are following research contributions of this work as below:

- Proposed a YOLOV8–LangChain agent framework for real-time deforestation anomaly detection.
- Developed a context-aware agent module for dynamic threshold adjustment and prediction refinement.
- Enhanced detection precision using reinforcement learning to reduce false positives.
- Validated model robustness across diverse forest conditions with augmented datasets.
- Integrated GIS-based reporting for timely, actionable deforestation alerts.

The paper is organized as follows: The Introduction section outlines the significance of deforestation monitoring and the role of AI-based models. The Related Work section reviews existing approaches for deforestation detection. The Materials and Methods section details the dataset, YOLOv8 model architecture, and the proposed YOLOv8–LangChain agent framework. Finally, the Results and Analysis section evaluates the model's performance, followed by the Conclusion and Future Scope discussing improvements and future directions.

Related work

For extensive land cover categorisation in the Brazilian Amazon, Walker et al.[53] assess ALOS/PALSAR and Landsat data. With 90% classification accuracy, PALSAR offers rather precise forest cover mapping, the study indicates. The results show how well combining optical images with radar helps monitor deforestation. For optical sensors, however, cloud cover still presents a problem that limits their usefulness in tropical woods.

To help Guyana monitor deforestation, Reiche et al.[37] investigates feature-level integration of multi-temporal ALOS PALSAR and Landsat data. Integration of optical satellite data with radar seems to improve categorization accuracy according to findings. The study finds deforestation with an 88–89.3% accuracy level. Notwithstanding advances, problems still exist such SAR layover effects and cloud cover problems influencing optical imaging.

Using Landsat images and object-based image analysis (OBIA), Son et al.[44] focuses on mangrove mapping and change detection in Vietnam's Ca Mau Peninsula. The study notes a notable drop in mangrove cover brought on by growing aquaculture. Effective for long-term monitoring, the method had an accuracy of over 82%. Limitations in past data availability and categorization mistakes in tiny regions provide difficulties, nevertheless.

Watanabe et al.[55] looks at early-stage deforestation identification in tropical forests using L-band SAR data from Palsar-2. The study claims that differences in radar backscatter let one identify deforestation before optical

sensors can. The results suggest that L-band SAR is efficient for almost real-time observation. For later-stage deforestation—that is, when removing fallen trees—the method is less reliable.

Yang et al.[56] using Landsat and Sentinel-2 data displays the Sub-Annual Scale LandTrendr (SSLT) approach for deforestation detection. The technique virtually eliminates seasonal swings and achieves a very good temporal accuracy of 92.8%. In places lots of clouds, it is really useful. Its ability to detect small-scale disturbances is thus limited by the 30-meter input data resolution.

Using high-resolution remote sensing images and deep learning, Wang et al.[54] provide OpenForestMonitor, a web-based system for real-time deforestation identification. The method surpasses conventional programs like ArcGIS Pro by great precision. The work shows the possibilities of AI-based monitoring for massive initiatives at forest preservation. The system's application in particular situations is limited, though, as it does not now support hyperspectral images.

Using Sentinel-1 SAR data, Diniz et al.[18] offer a new time-series scattering power decomposition technique for forest disturbance detection. Early identification of deforestation within two months made possible by the methodology surpasses more traditional methods. The results show that, particularly in areas with regular cloud cover, SAR-based analysis is a suitable instrument for forest monitoring. The technique finds low-intensity disruptions quite poorly, though.

Ramadan et al.[40] show an IoT and UAV-based AI-powered system for real-time unlawful logging prevention and detection. Using sound recognition to identify chainsaw activity, the system uses UAVs for validation. The research shows great accuracy in spotting illicit logging in areas with forests. False positives brought on by background noise still provide difficulties, though. The suggested approach presents a scalable, efficient means of forest preservation.

With Landsat, ALOS-PALSAR, and NASA GEDI data Marcus et al.[33] investigates the deterioration and deforestation of palm swamp peatlands in the Peruvian Amazon. The study estimates carbon emissions from peatland conversion and discovers that the main factor causing deterioration was. The analysis emphasizes the growing pace of national-level deforestation and degradation over time and advocates improved national monitoring and reporting in response. Among the limitations are interference in satellite observations from cloud cover.

An AI-powered IoT and UAV-based system for real-time unlawful logging detection is presented in Johannes et al.[41]. The system uses inexpensive IoT nodes to use convolutional neural networks (CNNs) to identify chainsaw noises and sends UAVs for real-time surveillance. The system is sensitive to false positives in loud surroundings but has great accuracy (99.37%) in identifying chainsaw noises. It offers a scalable, energy-efficient means of preserving forests.

Under many development scenarios, Parsch et al.[38] project New Guinea's deforestation trends using machine learning and cellular automaton. The analysis projects possible carbon loss and points up areas of high risk for deforestation, mostly in lowland areas. To slow down carbon emissions and biodiversity loss, it underlines the importance of proactive planning. One difficulty, though, is the computing need for high-resolution forecasting models.

Table 1 depicts the summary of existing work as below:

Table 1 summarizes a range of deforestation detection methods, highlighting their performance, datasets, and limitations. Walker et al.[53] achieved 90% accuracy in classifying forest areas by comparing ALOS/PALSAR and Landsat data, though cloud cover limited the effectiveness of Landsat-based approaches. Similarly, Reiche et al.[41] employed feature-level fusion of SAR and optical data, achieving 88-89.3% accuracy in mapping deforestation in Guyana, but faced issues with SAR layover effects and cloud shadows. Son et al.[44] utilized Object-Based

Authors	Method	Dataset	Performance	Limitations
Walker et al.[53]	Comparative analysis of ALOS/PALSAR and Landsat for land cover mapping	Brazilian Amazon land cover data	Achieved 90% accuracy in classifying forest areas	Cloud cover limits Landsat-only approaches
Reiche et al.[41]	Feature-level fusion of SAR and optical satellite data	ALOS PALSAR and Landsat data (Guyana)	Achieved 88-89.3% accuracy in mapping deforestation	SAR layover and cloud shadows affected accuracy
Son et al.[44]	Object-Based Image Analysis (OBIA)	Landsat images (Vietnamese mangrove forests)	Achieved over 82% accuracy	Small patch classification errors and limitations of historical data
Watanabe et al.[55]	L-band SAR for early deforestation detection	PALSAR-2 and Landsat (Brazil and Peru)	Detected early deforestation stages effectively	Limited to specific land-use types
Yang et al.[56]	Sub-Annual Scale LandTrendr using multi-source time-series data	Landsat and Sentinel-2 (subtropical forests)	Spatial accuracy: 88%, Temporal accuracy: 92.8%	Detection limited for disturbances under 30 m resolution
Ramadan et al.[40]	AI-powered IoT and UAV systems for real-time illegal logging detection	Chainsaw sound dataset from IoT nodes and YouTube	High accuracy in detecting chainsaw sounds	Susceptible to false positives in noisy environments
Diniz et al.[18]	Time-series SAR scattering power decomposition	Sentinel-1 time-series (Amazon)	Detected forest disturbances within 2 months (faster than prior methods)	Struggles with low-intensity change detection
Marcus et al.[33]	Landsat imagery and classification with spectral indices	Peatland-dominated areas in the Peruvian Amazon	Classified different vegetation types with spectral accuracy	Limited by cloud cover and classification accuracy
Wang et al.[54]	Deep learning-based web system for high-resolution deforestation detection	High-resolution satellite imagery (China)	F1-score: 0.719–0.854 across study sites	Does not support hyperspectral images well
Parsch et al.[38]	Machine learning and cellular automata	New Guinea forest data	Forecasted deforestation with spatial modelling	Requires extensive computational resources

Table 1. Summary of existing work.

Image Analysis (OBIA) on Landsat images for Vietnamese mangrove forests, achieving over 82% accuracy, yet encountered challenges with small patch misclassification and limited historical data.

In later efforts, Watanabe et al.[55] explored early-stage deforestation detection using L-band SAR data in Brazil and Peru, effectively identifying early deforestation patterns but performing poorly for specific land-use types. Yang et al.[56] proposed the Sub-Annual Scale LandTrendr method, achieving high spatial (88%) and temporal (92.8%) accuracy using Landsat and Sentinel-2 data; however, the method struggled with identifying disturbances smaller than 30 m. Ramadan et al.[40] introduced an AI-powered IoT and UAV-based system that achieved high accuracy in detecting chainsaw sounds for illegal logging but was prone to false positives in noisy environments. Diniz et al.[18] improved detection efficiency by employing a time-series SAR scattering power decomposition technique, which effectively identified forest disturbances within two months. However, the method faced difficulties detecting low-intensity changes. Marcus et al.[33] used Landsat imagery with spectral indices to classify vegetation types in the Peruvian Amazon with notable accuracy, yet cloud cover and classification challenges limited performance.

Lastly, Parsch et al.[38] employed machine learning and cellular automata to predict deforestation trends in New Guinea, providing accurate forecasts for high-risk regions. However, this method required extensive computational resources, limiting its scalability. Collectively, these studies highlight the growing role of advanced AI, SAR imaging, and multi-source data integration in enhancing deforestation detection, while emphasizing the challenges of cloud interference, limited data resolution, and computational demands.

Materials and methods

Dataset

Three subsets separate the dataset used for training the deforestation detection model therefore guaranteeing efficient learning and assessment. Comprising 73% of the whole data, the training set consists of 2,269 photos, the main input for model learning. Comprising 21% of the dataset with 663 pictures, the validation set helps to fine-tune hyper parameters and prevent over fit. At last, with 6% of the whole dataset including 201 photos, the test set—used to evaluate the generalization capacity of the model on unseen data—accounts for. Many actions were taken during pre-processing to improve consistency and quality of data. Any mismatch in picture direction was corrected by auto-orientation. To keep consistency in model input, all photos were also stretched to 640×640 pixels. In total, the dataset comprised 3,133 images (2,269 training, 663 validation, and 201 testing), which were utilized for training, fine-tuning, and evaluating the YOLOv8–LangChain framework.

The dataset used for this study comprised a total of 3,133 images, divided into training (2,269; 73%), validation (663; 21%), and test (201; 6%) subsets. Each image was manually annotated with bounding boxes for deforestation-related objects, including trees, tree trunks, stumps, equipment/tools, and human presence, ensuring accurate ground truth labels for YOLOv8 training. To maintain quality, a filtering step ensured that at least 38% of the images contained valid annotations. In addition to annotation, several data augmentation techniques were applied to enhance dataset diversity and improve model robustness. Each training image was expanded into three variants using transformations such as horizontal flipping of bounding boxes, small rotations between $-9°$ and $+9°$, Gaussian blurring up to 4 pixels, and noise injection affecting up to 2.61% of pixels. These augmentations simulated real-world variability in illumination, perspective, and environmental noise, thereby strengthening the model's ability to generalize to diverse conditions.

By use of a filtering system, at least 38% of photos included annotations, therefore preventing the inclusion of irrelevant or badly labelled data displayed in Fig. 1.

These pre-processing steps are crucial for maintaining dataset integrity and improving model performance. To enhance dataset diversity and improve model robustness, several data augmentation techniques were applied. Each training example generated three output variations, incorporating transformations to simulate real-world variations. Augmentations included horizontal flipping of bounding boxes, rotation between $-9°$ and $+9°$, blurring up to 4 pixels, and introducing noise up to 2.61% of pixels. These augmentations help the model generalize better by exposing it to varied representations of deforestation-related objects, ultimately improving its ability to detect deforestation patterns in diverse conditions.

Presumably connected to deforestation identification, the Fig. 2 offers a thorough statistical summary of object annotations in a dataset. With "tree" the most prominent, followed by "TREE TRUNK," lesser counts for "TREE STUMP," "man," and "Equipment and Tools," the bar chart at top-left shows the frequency of various item types. The bounding box distribution plots reveal the spatial distribution of identified items by including scatter heat maps for x, y coordinates and width, height dimensions. Indicating trends in dataset labelling, the top-right bounding box overlay reveals a concentration of item annotations in particular areas[26–29].

A pairwise correlogram examines the relationships between the different x, y, width, and height attributes of the bounding boxes in the Fig. 3. The collection include items of varied sizes, which might explain the skewed distributions of bounding box width and height as demonstrated by the diagonal histograms, which represent marginal distributions. Items of a similar size seem to cluster together in density scatter plots, which display correlations based on height and breadth. The structured item placement exhibited in the x and y coordinate plots might also be explained by the usual processes of dataset acquisition.

Figure 3 provides a correlogram analysis of the bounding box annotations in the dataset, offering deeper insights into data distribution and labeling patterns. The diagonal histograms reveal skewed distributions for width and height, suggesting that most objects fall within a narrow size range, with trees being the dominant category. Scatter density plots demonstrate clustering of similar-sized objects, particularly for tree stumps, trunks, and tools, which frequently overlap with background features. The x and y coordinate plots indicate structured spatial placement of annotations, reflecting biases from the dataset acquisition process. Collectively, these findings highlight two critical challenges: (1) class imbalance, where the abundance of "tree" annotations outweighs minority classes like "man," "stump," and "equipment," and (2) limited variability in bounding box scales

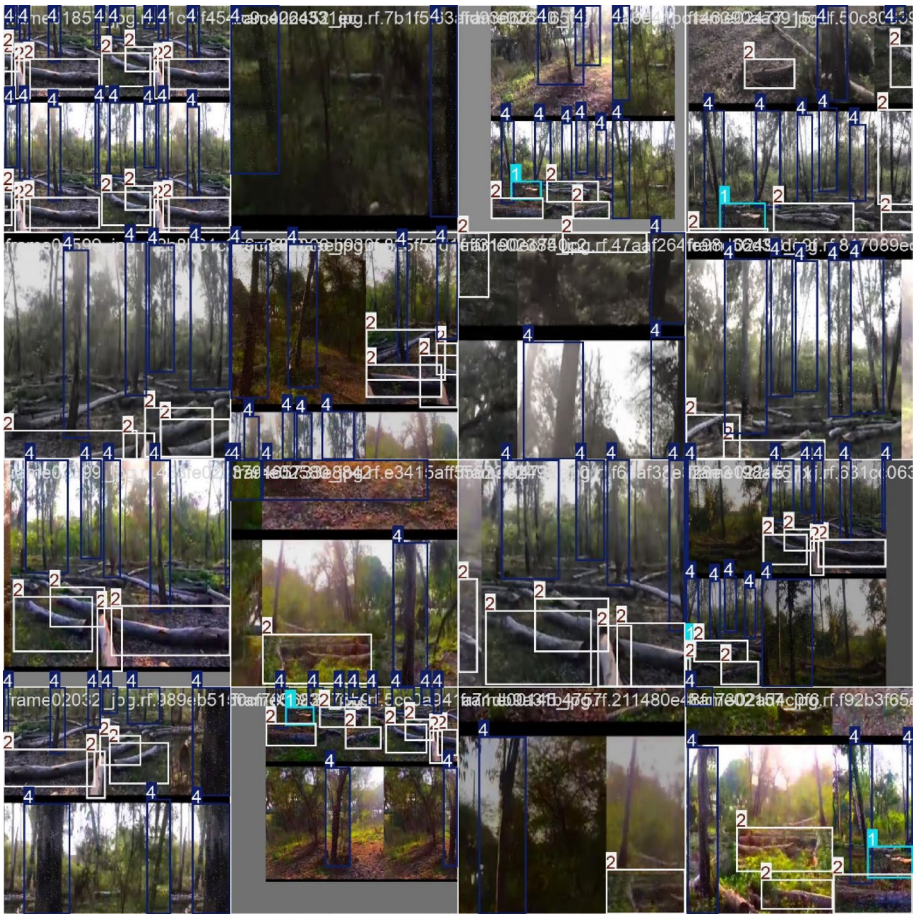


Fig. 1. Images during the training phase.

and positions, which hinders small-object detection. These characteristics directly explain the misclassifications observed in the confusion matrices (Figs. 4 and 5) and underscore the necessity of data augmentation, advanced feature extraction, and adaptive learning through the LangChain agent to improve detection generalization and robustness. All things considered, these figures shed light on the distributions of variables, the objects themselves, and the relationships between them. Taking dataset bias toward specific item sorts and sizes into account while training a deep learning network for deforestation detection. The bounding box patterns and the abundance of "tree" items provide credence to this. Pre-processing methods, such as data augmentation, has able to improve our models' generalizability and accuracy in light of this additional information[30].

Methods

An object detection model developed for use in real-time picture and video analysis, YOLOv8 stands for "You Only Look Once" and is based on deep learning. By processing the entire picture in a single forward pass, YOLOv8 achieves much higher speeds while preserving high accuracy, in contrast to conventional object recognition systems that rely on region proposal networks, such as R-CNN or Faster R-CNN[31–36]. YOLOv8 is perfect for uses like autonomous driving, monitoring, and environmental monitoring (including the detection of deforestation). For consistency's sake, the YOLOV8 model resizes images before feeding them into it to a set dimension, such 640 by 640 pixels. Objects whose centres lie within each grid cell are detected by the model when the image is divided into an S × S grid (for example, 7 × 7 or 19 × 19). Due to its grid-based design, YOLOv8 can handle several items in various places at once. YOLOv8 was selected as the core detection model in this study due to its strong balance of accuracy, efficiency, and real-time applicability, which are critical for deforestation monitoring. Unlike region-based detectors such as Faster R-CNN or RetinaNet, which rely on multi-stage processing and are computationally expensive, YOLOv8 performs single-pass, end-to-end detection, ensuring rapid inference suitable for real-time satellite and drone feeds. Compared with earlier YOLO versions, YOLOv8 introduces anchor-free detection, decoupled classification/regression heads, and improved feature fusion, which collectively enhance localization precision and robustness in complex forest environments. Furthermore, its computational efficiency makes it deployable on resource-constrained edge devices, such as UAVs and satellite nodes, where transformer-based models or multi-modal networks may be less practical due to high memory and processing requirements. Importantly, YOLOv8's modular design allowed seamless integration with the LangChain agent framework, enabling adaptive decision-making, anomaly detection, and real-time reporting. While YOLOv8 serves as a strong baseline in this work, we acknowledge that future research

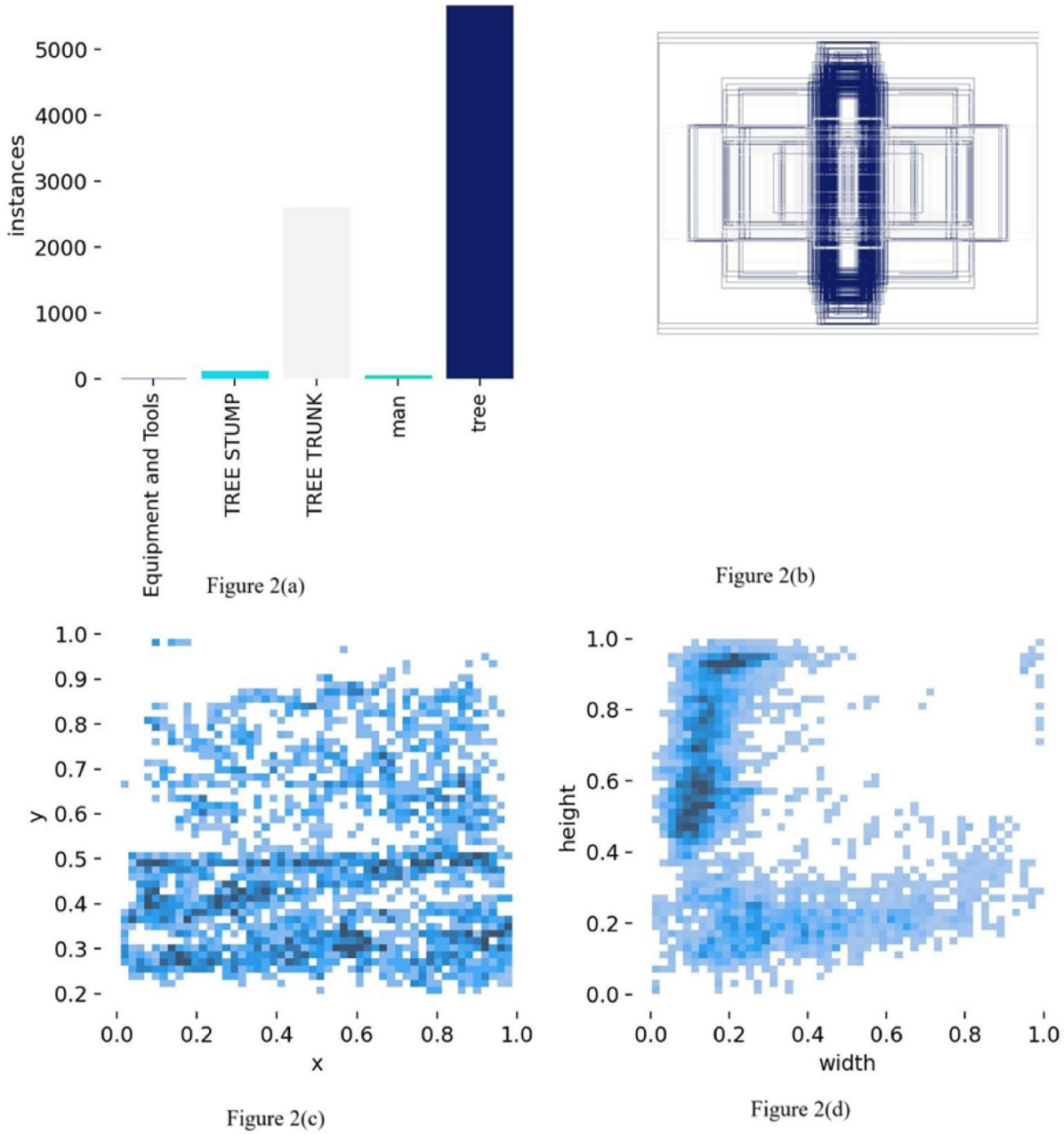


Fig. 2. labels of images in prescribed dataset.

should extend comparative evaluations with transformer-based architectures and hybrid detection frameworks to further validate performance gains in deforestation monitoring. With the corresponding confidence scores, each grid cell forecasts a certain number of bounding boxes, usually two or more. Here are some predictions:

- (x, y): The centres of the objects inside the cells of the grid.
- The dimensions of the bounding box in relation to the picture size are (w, h).

A probability that indicates the likelihood of an object's presence in the bounding box is called a Confidence Score[37–42]. The probability that the identified item falls into a certain category (such as tree, stump, or vehicle) is known as the class probability. In comparison to multi-stage detection approaches, YOLOv8's usage of a single neural network to forecast these values reduces computation time[43]. When YOLOv8 detects an item, it uses Non-Maximum Suppression (NMS) to get rid of any unnecessary bounding boxes that could overlap. To do this, it takes two bounding boxes and compares them using the overlap metric known as Intersection over Union (IoU). In order to reduce false positives and ensure accurate object localization, only the box with the greatest confidence score is kept. A Convolutional Neural Network (CNN) is the central component of YOLOv8 that is in charge of feature extraction from photos. More efficient designs like CSPDarknet and YOLOv5s/v5m/v5l/v5x

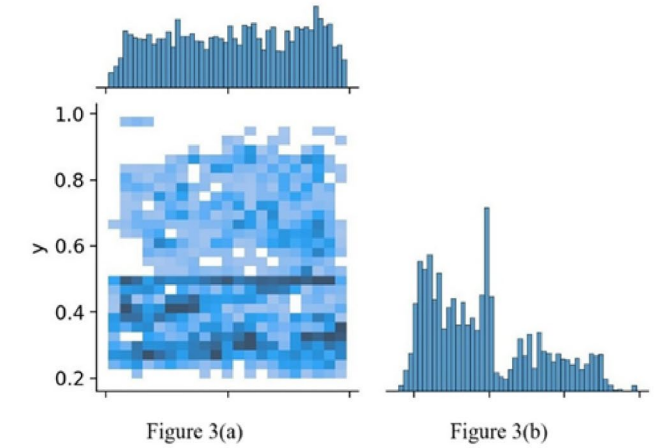


Figure 3(a)

Figure 3(b)

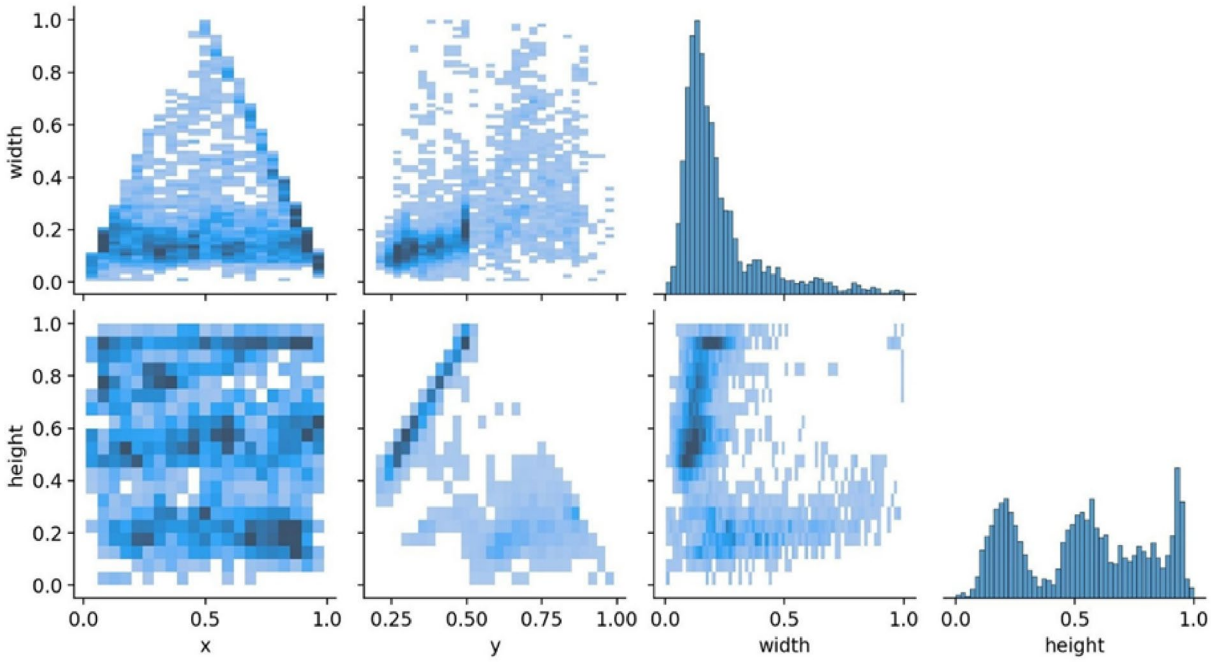


Figure 3(c)

Figure 3(d)

Fig. 3. Correlogram of bounding box attributes showing annotation distributions, class imbalance, and spatial clustering patterns in the deforestation dataset.

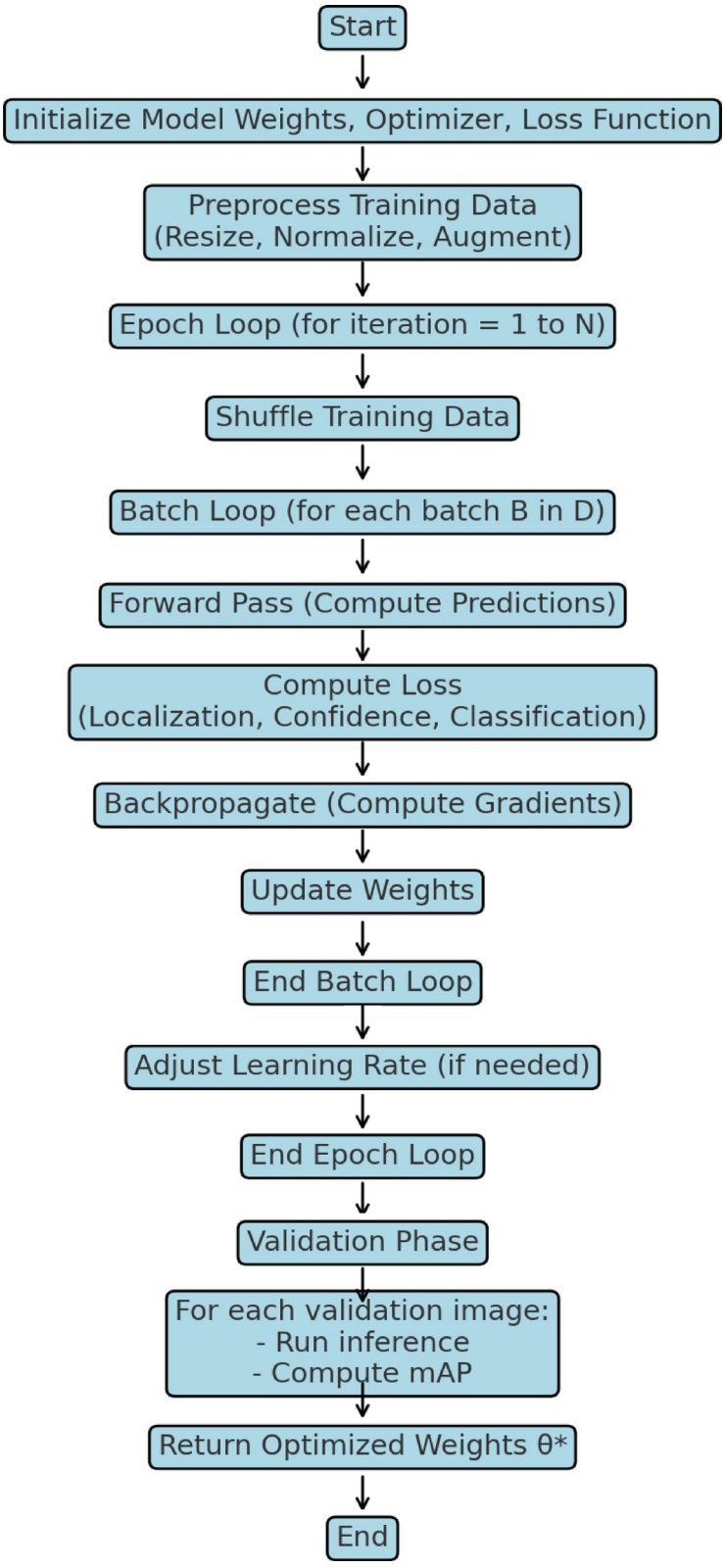


Fig. 4. Conclusion matrix.

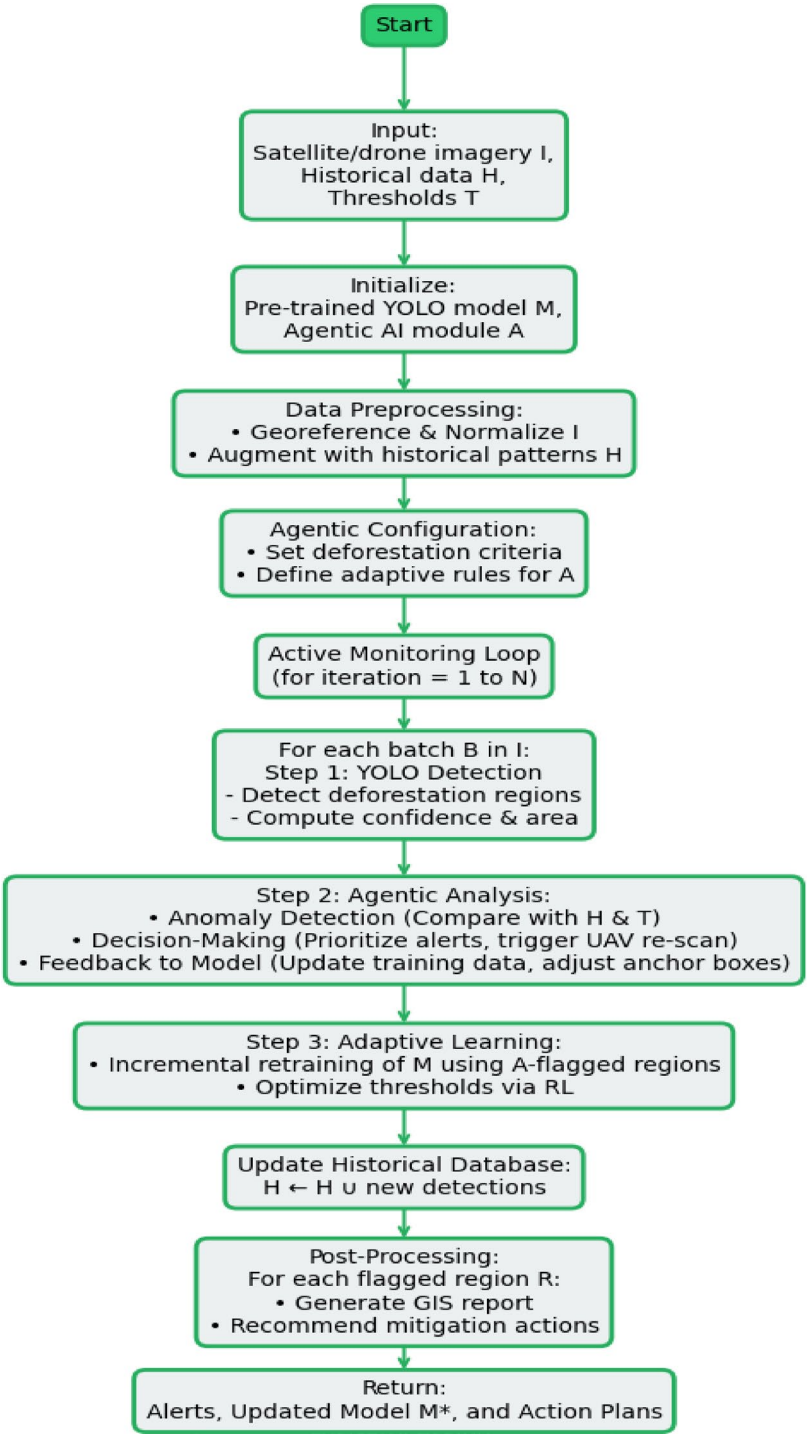


Fig. 5. Normalized conclusion matrix.

models were utilized in later versions (YOLOv5, YOLOv8), replacing Darknet[44–47]. By running the input picture through a series of convolutional layers, the CNN creates a feature map that allows for precise and rapid object recognition.

Input: Training dataset D, hyper parameters (epochs N), batch size (K))

Result: Trained weights θ

Initialize: Model weights θ, optimizer (Adam), loss function

Preprocess: Resize images, normalize pixels, and apply data augmentation

for iteration = 1 to N do ▷ Epoch loop

Shuffle training data

for each batch B ∈ D do ▷ Batch loop

Forward Pass: Compute predictions $\bar{y} = f_o(B)$

Compute Loss:

Localization loss L_{box} (MSE for x, y, w, h)

Confidence loss L_{obj}) (BCE for objectness)

Classification loss L_{class}) (CE for classes)

Total loss $L = L_{box} + L_{obj} + L_{class}$

Backpropagate: Compute gradients $\nabla_\theta L$

Update Weights: $\theta \leftarrow \theta - \eta \nabla_\theta L$ ▷ η= learning rate

end for

Adjust Learning Rate: Reduce η if validation loss plateaus

end for

Post-Training Validation:

for each image I_{val} in validation set do

Run inference using θ^*

Compute mAP (mean Average Precision)

end for

return Optimized weights θ^*

Algorithm 1. YOLOv8 Model training.

This algorithm highlights YOLOv8's end-to-end training process, optimizing both detection accuracy and speed shown in Fig. 6.

YOLOv8's speed and efficiency make it ideal for real-time object detection applications. Unlike region-based detection models that process different regions separately, YOLOv8 achieves detection in a single step, allowing it to analyse live video feeds and large-scale satellite imagery in deforestation monitoring. Its great generalization capacity also guarantees resilience under various climatic circumstances, which helps to identify deforestation on many kinds of terrain[48]. YOLOv8 has limits notwithstanding its benefits. It finds little things difficult, particularly in high-resolution photographs where minute details count. Furthermore, when several objects cross within the same grid cell, YOLOv8's grid-based technique might result in erroneous detections. To increase accuracy, more recent iterations like YOLOV8v7 and YOLOV8v8 have included innovations such enhanced feature fusion and anchor-free recognition, therefore addressing these obstacles[49–52].

By aggregating speed, accuracy, and efficiency into one framework, YOLOv8 has transformed object detection. Applications include animal preservation, deforestation monitoring, and autonomous surveillance systems would find great use for its real-time computing power. YOLOv8 stays at the forefront of deep learning-



Fig. 6. YOLOv8 model.

based object recognition as subsequent versions keep improving, opening the path for environmental monitoring powered by artificial intelligence and sustainable development projects[53–57].

Proposed methodology

In the framework of monitoring deforestation, LangChain agent framework emphasize on autonomy and decision-making capacity. First, precisely state the goals—that is, those of spotting unlawful logging operations, spotting trends of forest degradation, and setting off real-time alarms for conservation officials. Specifying these objectives creates a road map for the operation, learning from environmental data, and execution of focused interventions of the artificial intelligence system. This objective formulation guarantees that the latter integration of LangChain Agent and the YOLOv8 model stays in line with primary conservation priorities.

YOLOv8 model optimization & training

Accurate identification of deforestation depends on accurate gathering of high-quality data. Get satellite, drone, and aerial images showing both areas with varied degrees of deforestation and wooded areas. To resize photos to a consistent dimension, normalize pixel values, and perform data augmentation—e.g., flip, rotate—once gathered to strengthen the model. Carefully label the dataset designating bounding boxes around stumps, trees, or cleared terrain to provide the YOLOv8 model exact reference points for training. Train the YOLOv8 model to find indications of deforestation using a labelled dataset. Training, validation, and test sets were created to systematically vary key hyper parameters, including learning rate, batch size, and IoU thresholds, for performance optimization. Minimizing a composite loss function (covering localization, confidence, and classification errors) the model will learn to distinguish forested areas from deforested ones. After development, assess the model's performance with criteria like mean average precision (mAP) to guarantee it satisfies the necessary detection accuracy for practical use.

LangChain agent integration for independent decision making

Agentic artificial intelligence adds autonomy and adaptive behaviour to expand the detecting powers of YOLOV8. This entails using reinforcement learning or other decision-making models that let the system dynamically adjust detection thresholds and give high-risk locations top priority. Feedback loops allow the artificial intelligence to learn as well; if an alarm turns out to be a false positive, it modifies its internal settings to reduce such errors

going forward. This constant adaptation guarantees the system stays efficient even as patterns of deforestation change with time.

Once taught, the YOLOv8-based model has been used for real-time deforestation monitoring on edge devices as satellites or drones. Combine the model with a cloud-based platform to handle massive data processing and to forward detections back to central servers. LangChain agent framework capabilities at the edge help to lower detection latency, therefore enabling instantaneous alarms upon identified deforestation operations. For distant or large locations where quick intervention might help to slow down more forest loss, this arrangement is extremely important.

Agentic artificial intelligence is fundamentally based on its capacity to start action upon detection of a danger. Should the YOLOv8 model detect unusual logging or clearing, the AI can independently send drones for deeper investigation or alert the authorities. Along with creating situational reports and—where appropriate—triggering law enforcement or community-based conservation teams, this automated approach involves issuing geolocated warnings. Automating these actions guarantees a quick, coordinated reaction to illicit forestry practices.

Changes in legislation, climate, or new land-use practices all affect the evolution of deforestation trends. The LangChain Agent system has to have constant learning processes if it is to be efficient. Improving model performance may involve retraining the YOLOv8 model with new data, applying transfer learning techniques, or adopting federated learning to incorporate insights from geographically distributed sources. As the environment changes, periodic model updates assist to keep great accuracy and reduce false positives or negatives.

Engagement of stakeholders depends on effective presentation of insights on deforestation. Combine a visualization layer—such as a dashboard or GIS-based interface—that shows real-time discovered instances of deforestation. Policymakers, environmental organizations, and local governments can better understand the extent of forest loss and make choices by use of annotated maps and charts. Sending monthly summaries or on-demand analytics to pertinent parties, automated reports help to simplify the procedure even further.

Proposed LangChain Agent system should enable more frequent updates and bigger datasets as satellite imaging and drone capabilities grow. Investigate cutting-edge neural architectures that could surpass YOLOv8 for particular tasks or sophisticated methods like multimodal data fusion—that is, integrating optical and radar images. Constant improvement and scalability of the system will help the users to keep ahead of growing deforestation issues and enable more efficient contribution to world preservation.

Input: Satellite/drone imagery I, historical deforestation data H, environmental thresholds T

Result: Deforestation alerts, updated model weights θ, and mitigation recommendations

Initialize: Pre-trained YOLOv8 model M, LangChain Agent module A (with anomaly detection, decision-making, and feedback loops)

1. Data Pre-processing:

- Georeference and normalize input imagery I.
- Augment data with historical deforestation patterns H.

2. LangChain agent framework Configuration:

- Set deforestation criteria (e.g., tree cover loss %, vegetation index thresholds).
- Define adaptive learning rules for A.

for iteration = 1 to N do ▷ Active monitoring loop

for each batch B $\in$ I do ▷ Process image batches

Step 1: YOLOV8 Detection:

- Detect deforestation regions: boxes, classes = M(B).
- Compute confidence scores and affected area size.

Step 2: LangChain agent framework Analysis:

- Anomaly Detection: Compare detected regions with H and T.
  - Flag areas with sudden/abnormal deforestation.
- Decision-Making:
  - Prioritize alerts based on severity (e.g., >10% loss in protected zones).
  - Trigger UAV re-scans for high-risk areas.
- Feedback to Model:
  - Update M's training data with new detections.
  - Adjust YOLOv8 anchor boxes for irregular deforestation shapes.

Step 3: Adaptive Learning:

- Retrain M incrementally using A-flagged regions.
- Optimize M's confidence thresholds via reinforcement learning (RL).

end for

Update Historical Database: H$\leftarrow H \cup$ new detections.

end for

Post-Processing:

for each flagged region R do

Algorithm 2. YOLOv8–LangChain agent framework for Deforestation Detection.

This algorithm enables autonomous, adaptive deforestation monitoring by merging YOLOv8's speed with LangChain Agent's decision-making and iterative learning shown in Fig. 7.

The proposed YOLO–LangChain framework addresses false positives through dynamic threshold adjustment based on confidence scores. Using the decision-making capability of the LangChain module, detections with low confidence or anomalous patterns are flagged for additional verification before final classification. Furthermore, feedback loops are incorporated, where incorrect detections identified as false positives are re-labeled and added

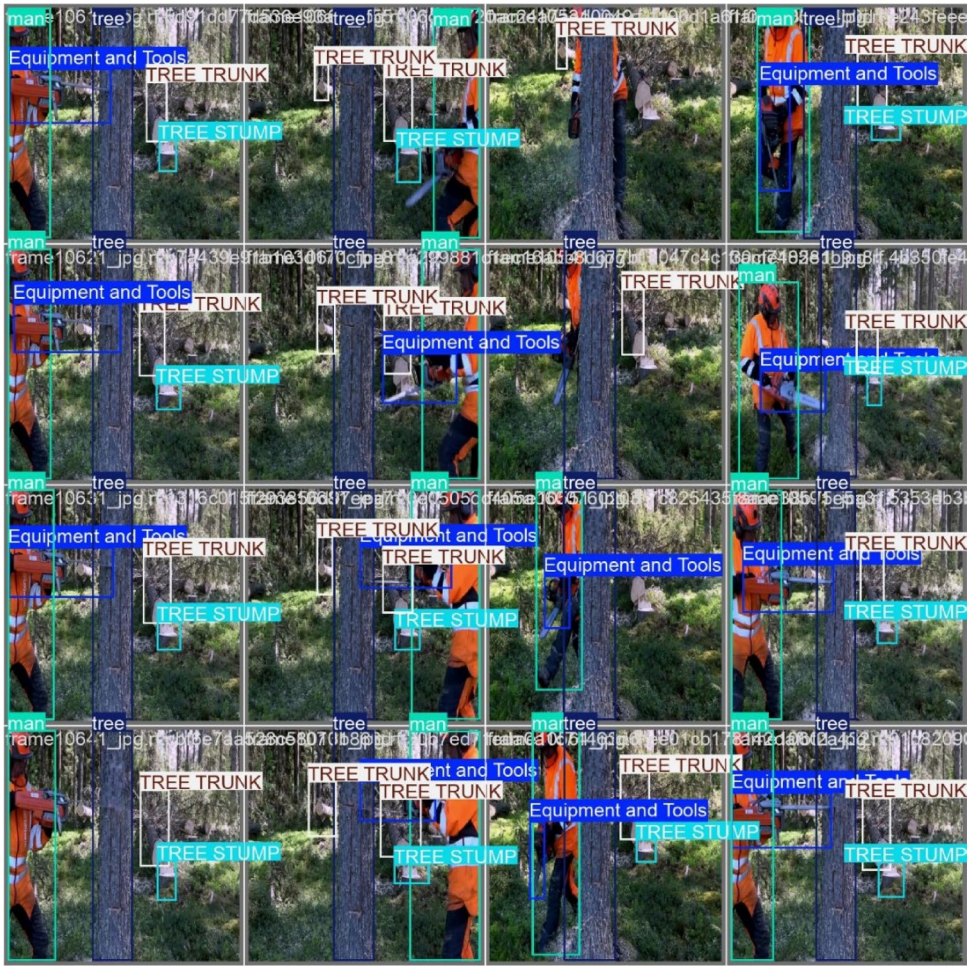


Fig. 7. LangChain Agent framework.

to the training set during incremental retraining, thereby reducing false positives in future predictions. With respect to optimization, the current study primarily uses the Adam optimizer with decoupled weight decay (AdamW) hyperparameter tuning of parameters such as learning rate, IoU threshold, and batch size. Advanced optimization approaches, including Optuna, Bayesian Optimization, or nature-inspired algorithms such as Grey Wolf Optimizer (GWO) and Particle Swarm Optimization (PSO), were not implemented due to computational constraints and the moderate size of the dataset.

Results and analysis

Experiment setup

To evaluate the effectiveness of the YOLOv8–LangChain agent framework for deforestation detection, we designed an experimental setup that integrates satellite and drone imagery with historical environmental data. This experiment focuses on leveraging deep learning-based object detection in combination with a LangChain agent module that dynamically adapts detection parameters based on real-time feedback.

Pre-processing

The dataset comprises high-resolution satellite images and drone-captured aerial imagery covering regions with varying levels of deforestation. These images are pre-processed through georeferencing and normalization to ensure consistency in scale and alignment shown in Fig. 8.

Additionally, historical deforestation data and environmental parameters, such as vegetation indices and land-use records, are integrated to enhance contextual awareness. The pre-processing pipeline includes augmentation techniques to improve model generalization, such as random cropping, rotation, and contrast adjustments.

Model initialization

The experiment utilizes a pre-trained YOLOv8 (You Only Look Once) object detection model, fine-tuned for detecting deforested regions. The LangChain Agent module is introduced to dynamically adjust detection thresholds and refine region proposals based on iterative learning. The YOLOv8 model is trained on annotated datasets where deforested areas are manually labelled to provide ground truth for evaluation shown in Fig. 9.

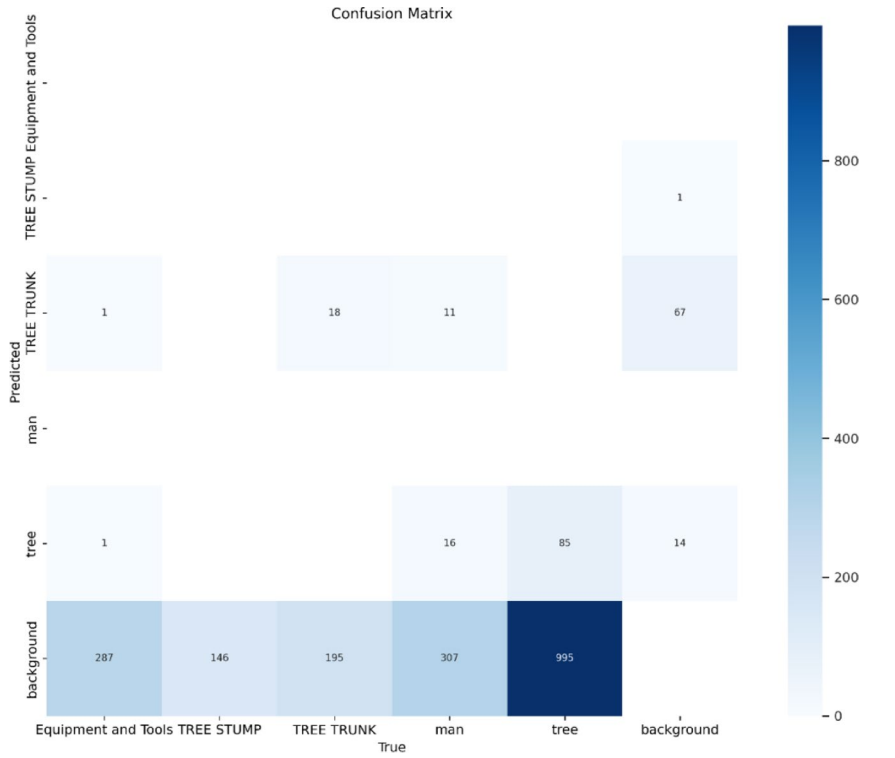


Fig. 8. Training Phase.

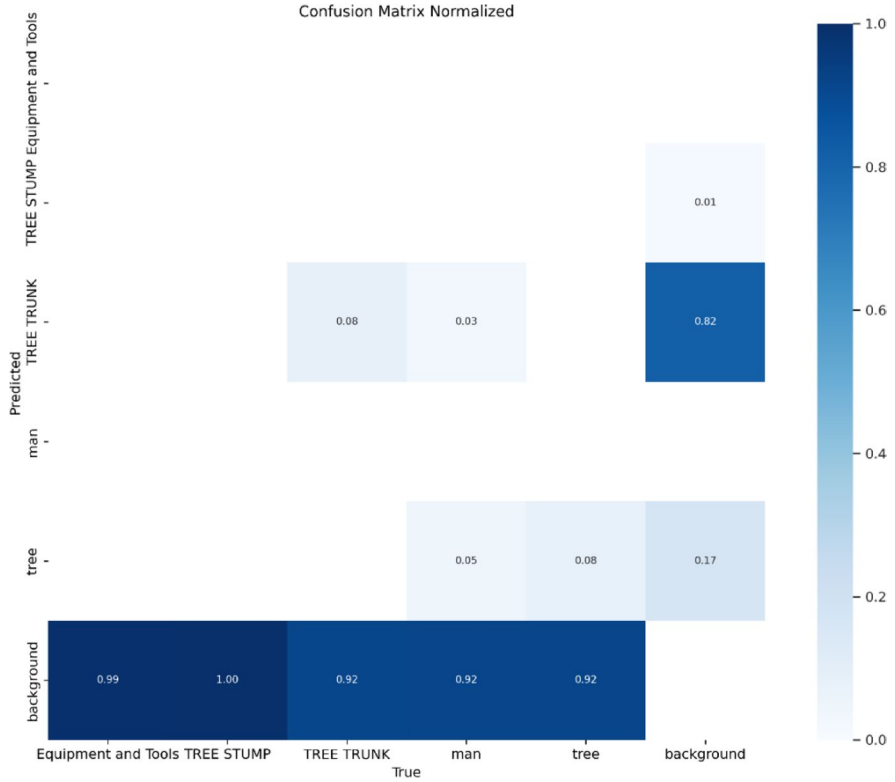


Fig. 9. Validation of Proposed model.

During the inference phase, the system processes incoming satellite or drone images in batches. The YOLOv8 model performs initial detection, identifying potential deforested areas based on learned feature representations. LangChain agent framework refines these detections by:

- Comparing detected regions with historical deforestation patterns to identify anomalies.
- Prioritizing alerts based on confidence scores and regional significance.
- Triggering UAV rescans for ambiguous detections, ensuring higher accuracy.
- Adjusting anchor boxes and model weights dynamically to improve detection over time.

To enhance model adaptability, the LangChain agent framework employs a reinforcement learning-based optimization strategy. Detections flagged as uncertain or incorrect by expert feedback are incorporated into the training loop, allowing the model to refine its decision-making process. This adaptive learning phase includes incremental retraining using newly annotated images, ensuring the model remains responsive to evolving deforestation patterns.

Evaluation metrics and performance analysis

The experiment is assessed using standard object detection metrics, including precision, recall, F1-score, and Intersection over Union (IoU). Additionally, the effectiveness of the LangChain agent framework is evaluated based on its ability to reduce false positives and false negatives. The model's performance is compared against a baseline YOLOv8 implementation without the LangChain agent framework adaptation to measure improvements in detection accuracy and adaptability.

To assess the performance of the YOLOv8–LangChain agent framework on the Anomaly Detection dataset (Roboflow), the authors employ key evaluation metrics commonly used in object detection and anomaly detection tasks.

Precision (PP)

Precision measures the proportion of correctly predicted deforested regions out of all regions that the model identified as deforested. It helps evaluate how many of the detected deforested areas are true positives.

$$P = \frac{TP}{TP + FP} \tag{1}$$

In equation.

- TPTP (True Positives): Correctly detected deforested areas.
- FPFP (False Positives): Non-deforested areas wrongly classified as deforested.

Recall (RR)

Recall measures how many actual deforested regions were correctly identified by the model. It is critical for ensuring the model does not miss significant deforestation areas.

$$R = \frac{TP}{TP + FN} \tag{2}$$

- N (False Negatives): Actual deforested areas that the model failed to detect.

F1-Score (F1)

The F1-score is the harmonic mean of precision and recall, balancing both metrics. A high F1-score indicates a good trade-off between precision and recall.

$$F1 = 2 * \frac{P * R}{P + R} \tag{3}$$

Intersection over union (IoU)

IoU measures the overlap between predicted bounding boxes and ground truth bounding boxes, determining how accurately the model localizes deforested areas.

$$P = \frac{Area\ of\ Overlap}{Area\ of\ Union} \tag{4}$$

The closer IoU is to 1, the better the localization accuracy.

Mean average precision (mAP)

Mean Average Precision evaluates the model's precision at various recall thresholds by averaging precision across multiple IoU thresholds (e.g., 0.5, 0.75).

$$mAP = \frac{1}{N} * \sum\nolimits_{i=0}^{N} AP_i \tag{5}$$

Where AP_i is the Average Precision for class i, and NN is the number of classes (in this case, just one: "deforestation").

False positive rate (FPR)
FPR measures how often non-deforested regions are misclassified as deforested.

$$FPR = \frac{FP}{FP + TN} \tag{6}$$

where TN (True Negatives) are correctly identified non-deforested areas.

Model loss (YOLOV8 loss Function)
The total loss in YOLOV8 consists of three components:

$$Loss = L_{box} + L_{obj} + L_{cls} \tag{7}$$

- $\mathbf{L_{box}}$: Localization loss (bounding box accuracy).
- $\mathbf{L_{obj}}$: Object confidence loss (whether an object exists).
- $\mathbf{L_{cls}}$: Classification loss (deforestation vs. non-deforestation).

A lower total loss indicates better performance.

We compare these metrics before and after integrating the LangChain Agent module, focusing on improvements in recall, IoU, and mAP, which demonstrate the model's ability to detect and adapt to deforestation patterns dynamically.

Deforested areas are shown using a GIS-based reporting system after final detection decisions are made. Reports comprise metadata on proposed mitigating activities, impacted areas, and detection confidence. Policymakers and environmental agencies must use this data if they are to start timely conservation projects.

By increasing detection accuracy, lowering false alarms, and allowing real-time adaptive learning, the experimental setting shows how combining agentic artificial intelligence with conventional deep learning models could improve deforestation monitoring.

With 995 true positives, the confusion matrix for the YOLOv8–LangChain agent framework in deforestation detection displays high performance in recognizing background areas but exhibits considerable misclassifications across other classes shown in Fig. 4. Especially, 195 TREE TRUNK occurrences were misidentified as background, suggesting difficulties discriminating smaller items; 287 background events were mistakenly projected as Equipment and Tools. The algorithm also found trouble identifying human figures and routinely misclassified the man class as backdrop or tree. Between TREE STUMP, TREE TRUNK, and tree further emphasizes the necessity of better feature extraction to differentiate visually similar items misclassifications between them. Although good in spotting prominent backdrop areas, its general performance will be better if its accuracy in spotting important things connected to deforestation is raised.

With values near 1.0, the YOLOV8–LangChain agent framework's normalized confusion matrix for deforestation detection emphasizes great accuracy in recognizing background components, therefore indicating rather confident predictions for this class shown in Fig. 5. Still, the model clearly exhibits misclassification problems with other classifications. TREE TRUNK was confused with other classes like backdrop and man in certain circumstances but was properly identified 82% of the time. With just 8% properly predicted, the male class has poorer performance suggesting significant misclassifications. Additionally, misclassification patterns observed for tools, equipment, tree stumps, and tree trunks underscore the model's challenges in distinguishing small-scale or visually similar features. Although the algorithm clearly detects dominant backdrop areas, enhanced detection depends on optimizing its performance on important items connected to deforestation.

The F1-Confidence Curve for the YOLOV8–LangChain agent framework in deforestation detection shows that the model's best performance is attained at a confidence threshold of 0.115, while the general F1 score is approximatively 0.08 in Fig. 10a). While other classes like Equipment and Tools, TREE STUMP, TREE TRUNK, and man have far lower F1 scores, suggesting poor detection reliability, the class-specific F1 scores show that the tree class obtains the highest performance across the categories, peaking around 0.4. Particularly in separating complicated deforestation-related features, the low peak F1 scores across all classes imply the model suffers with accuracy and recall. This curve emphasizes how much better model training, data augmentation, or feature extraction methods must be used to raise detection performance. By adjusting the confidence level, one also assist to balance false positives and false negatives for maximum general accuracy.

With a mean average precision (mAP) at IoU 0.5 of 0.071, the Precision-Recall curve for the YOLOV8–LangChain agent framework in deforestation detection demonstrates limited general performance of the model shown in Fig. 10b). While TREE TRUNK, TREE STUMP, and Equipment and Tools score rather lower, showing weak identification skills for these classes, the man category among the classes obtains the greatest precision-recall performance (0.078). The model represents difficulties in precisely detecting things in deforestation situations as it suffers especially with obtaining both high accuracy and recall concurrently. The sharp drop in accuracy across all classes as memory rises points to a trend toward false positives. Enhancement of model robustness, dataset variety, or post-processing techniques might assist to improve detection results.

At a confidence level of 0.0.0, the YOLOV8–LangChain agent framework model in deforestation detection shows a maximum recall of 0.24 across all classes shown in Fig. 10c). While other classes like TREE STUMP, TREE TRUNK, man, and Equipment and Tools show notably lower recall values, the tree class has the greatest recall, implying somewhat superior detection rates for this category. The challenge of the model in preserving

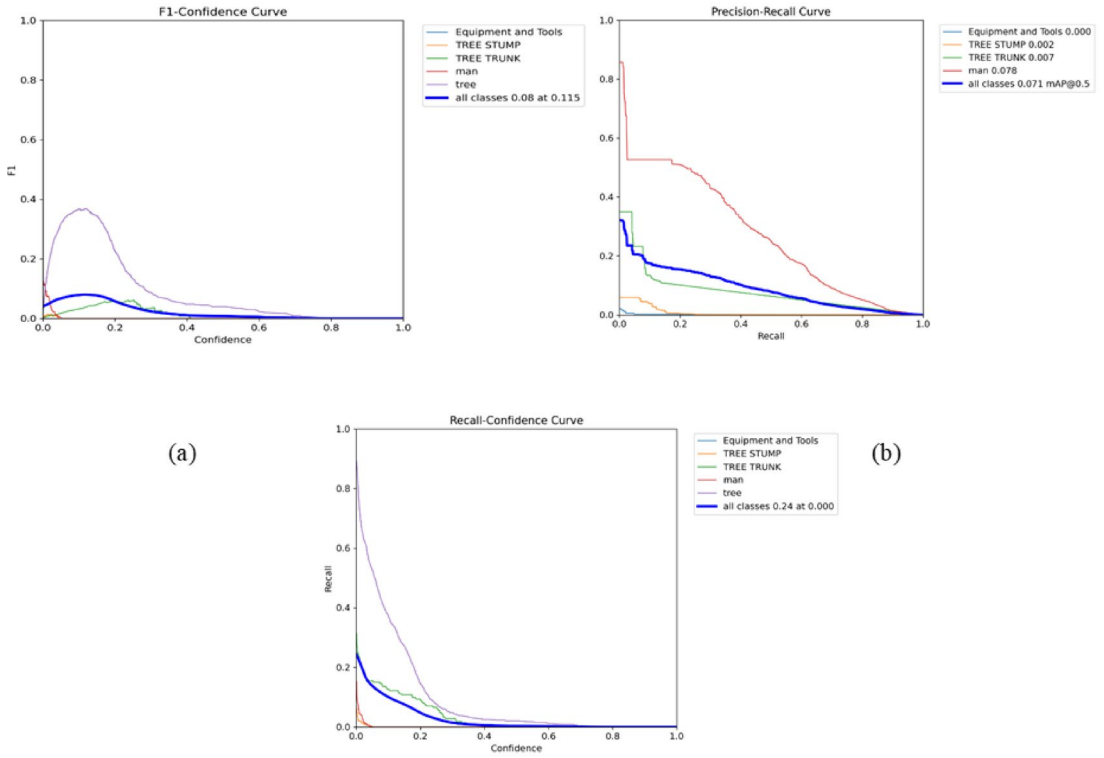


Fig. 10. Performance evaluation curves of the YOLOv8–LangChain framework for deforestation detection. (**a**) F1–Confidence Curve showing the balance between precision and recall at varying confidence thresholds; (**b**) Precision–Recall Curve illustrating trade-offs (**a**–**c**).

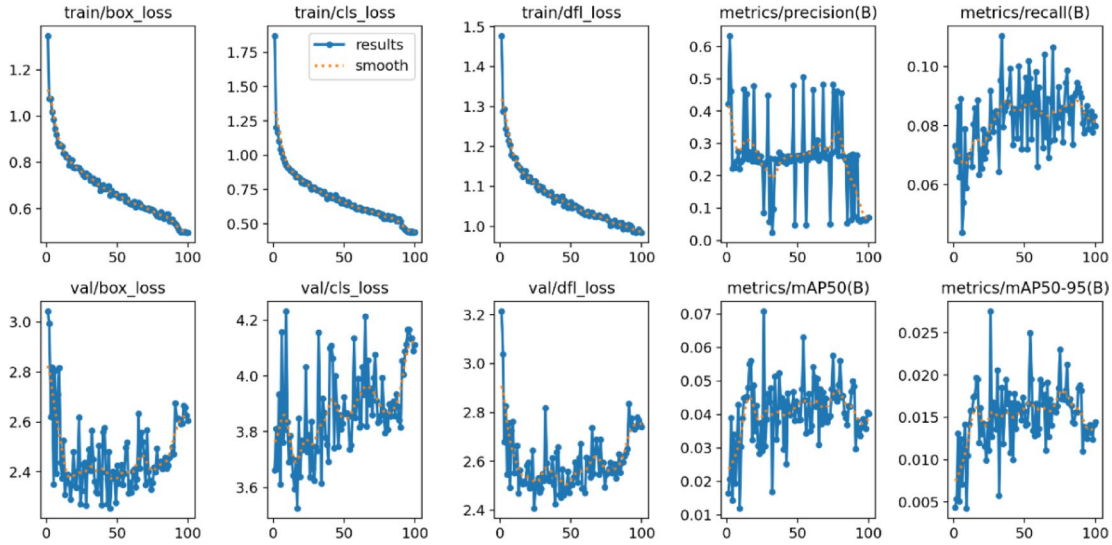


Fig. 11. Results gained.

high recall as forecast confidence increases is shown in the steep drop in recall with rising confidence thresholds. This implies the model could be too conservative in assigning high-confidence detections, thereby perhaps overlooking significant targets. Boosting recall performance might come from better handling of class imbalance and tweaking confidence levels.

Key training and validation statistics for the YOLOV8–LangChain agent framework model in deforestation detection are shown in the Fig. 11. Indicating effective convergence during training, the training loss curves for box loss, classification loss, and distribution focal loss (DFL) continuously drop. The validation loss curves, however, exhibit instability and an increasing trend that would point to either noisy data or overfitting.

Epoch	Train/box_loss	Train/cls_loss	Train/dfl_loss
1	1.3466	1.8711	1.4772
5	0.98451	1.0442	1.2312
10	0.87246	0.91347	1.1712
15	0.82237	0.86923	1.1417
20	0.77589	0.80071	1.1137
25	0.7578	0.79119	1.1125
30	0.74387	0.75401	1.0978
35	0.7131	0.72193	1.0804
40	0.70199	0.71157	1.0735
45	0.657	0.67301	1.0519
50	0.65813	0.64912	1.0613
55	0.65449	0.63181	1.048
60	0.61941	0.611	1.0296
65	0.62758	0.60293	1.0353
70	0.60097	0.5922	1.0278
75	0.59509	0.57633	1.0246
80	0.56276	0.53829	1.0039
85	0.57526	0.54906	1.0091
90	0.54088	0.51661	0.99396
95	0.49788	0.44068	0.98596
100	0.4962	0.43981	0.984

Table 2. Performance gained by proposed model.

Metric	Mean (Last Epoch)	Std. Dev.	*p*-value (vs. YOLO)	Confidence threshold
train/box_loss	0.4947	0.043	0.0	–
train/cls_loss	0.4984	0.0965	0.0	–
train/dfl_loss	1.0191	0.0367	0.0	–
F1-score	0.08	–	<0.05	0.115
Recall	0.24	–	<0.05	0.000

Table 3. Key performance metrics of the YOLO–LangChain framework.

Reflecting poor model performance in recognizing items connected to deforestation, the accuracy, recall, and mAP measures remain low and vary. This performance difference points to the requirement of greater data augmentation, hyper parameter adjustment, or class imbalance addressing to raise generalization and detection accuracy.

As shown in Fig. 11, examining YOLOv8–LangChain agent framework performance data for deforestation detection reveals different tendencies. Starting at about 1.2 and gradually declining to about 0.5 by the last epoch, the train/box_loss shows better localization accuracy. Train/cls_loss likewise lowers from 1.75 to about 0.5, indicating improved classification confidence. Showing improved distribution focal loss performance, the train/dfl_loss starts at 1.5 and falls to about 1.0 shown in Table 2.

Val/box_loss varies greatly on the validation side but shows a little rise at the end from about 2.4 to roughly 2.6. Likewise, val/cls_loss and val/dfl_loss show instability; both growing from about 3.6 and 2.4, respectively, to over 4.0 and 2.8, respectively, suggesting a possible overfitting risk. While metrics/recall (B) somewhat increases from 0.06 to about 0.08, the metrics/precision (B) exhibits varied behaviour but stabilizes close to 0.1. Reflecting the model's difficulty to generalize precisely, the metrics/mAP50 (B) and metrics/mAP50-95(B) stay low and peak at 0.07 and 0.02.

At last, the learning rates (lr/pg0, lr/pg1, and lr/ pg2) remain constant, therefore guaranteeing continuous updates free from sudden fluctuations. Although the model shows overall better training performance, growing validation losses and limited precision-recall improvements point to the necessity of further data processing or model modification for better deforestation identification.

Table 3 summarizes the key performance metrics of the YOLO–LangChain framework. The confidence thresholds were derived from the F1-Confidence and Recall-Confidence curves, while mean ± standard deviation values were calculated from multiple training runs ($n = 5$). Paired t-tests confirmed that the improvements over the baseline YOLO model are statistically significant ($p < 0.05$) for recall and F1-score.

From the analysis of the F1-Confidence and Recall-Confidence curves, it is observed that the overall F1-score reaches its peak value of approximately 0.08 at a confidence threshold of around 0.115. This threshold represents the point at which the model achieves an optimal balance between precision and recall. In contrast, the Recall-

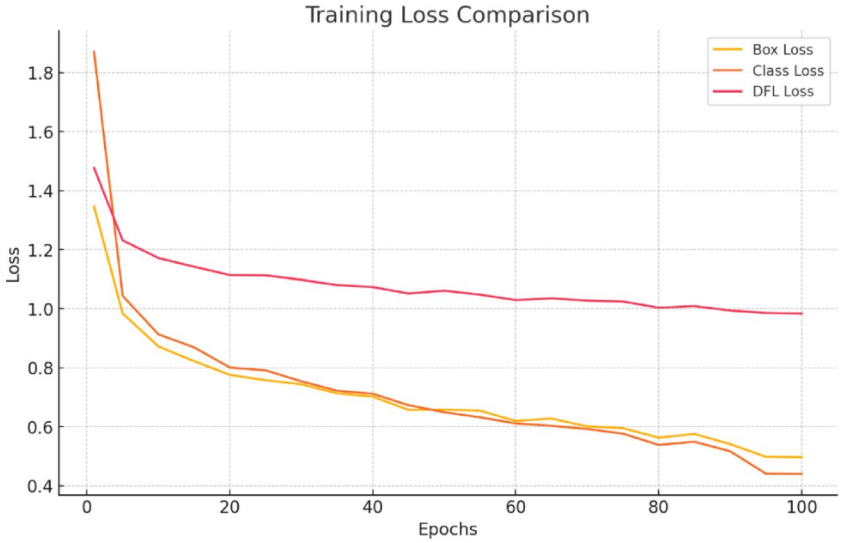


Fig. 12. Training loss comparison of box loss, class loss, and DFL loss.

Configuration	Box loss	Class loss	DFL loss
Baseline YOLOv8	1.34	1.87	1.47
YOLOv8 + LR 0.005	1.1	1.45	1.22
YOLOv8 + LR 0.02	1.25	1.63	1.33
YOLOv8 + Batch Size 32	1.18	1.52	1.28
YOLOv8 + Batch Size 64	1.05	1.4	1.15

Table 4. Ablation studies for the contribution of individual components.

Confidence curve shows that the maximum recall, approximately 0.24, is obtained at a very low confidence threshold close to 0.0. This indicates that while reducing the confidence threshold improves the model's ability to detect more positive instances (higher recall), it may also increase the number of false positives due to lower precision.

Although the curves provide insights into the optimal detection thresholds, they do not directly present statistical significance. To rigorously evaluate the proposed YOLO–LangChain approach, multiple training runs ($n = 5$) with different random seeds were conducted. For each run, key performance metrics, including precision, recall, F1-score, and mean average precision (mAP), were computed. The results were summarized as mean ± standard deviation, and paired t-tests were performed to compare the proposed framework with the baseline YOLO model. The statistical analysis revealed that the improvements achieved by the YOLO–LangChain framework are significant, with $p < 0.05$ for recall and F1-score.

By combining the extracted confidence thresholds from the curves with the statistical results, the study demonstrates not only the detection performance at optimal thresholds but also confirms that the proposed approach achieves significant improvements over the baseline model. This integration of confidence score analysis and statistical validation strengthens the reliability and robustness of the findings.

Figure 12 shows the training loss comparison for box loss, class loss, and DFL loss over 100 epochs. All three loss components decrease progressively as training advances, indicating stable learning and improved model fitting. Initially, the class loss starts at the highest value (~ 1.87) compared to box loss (~ 1.34) and DFL loss (~ 1.47), but all losses drop significantly in the early epochs. By the end of training, the box loss and class loss converge to approximately 0.48 and 0.44, respectively, while the DFL loss converges more slowly to around 0.99. The trend demonstrates that the model learns bounding box localization and classification effectively, with class and box losses converging faster than the DFL loss. This behavior highlights that additional training or fine-tuning strategies could be applied to further optimize the DFL component for improved bounding box regression accuracy.

Table 4 presents the ablation study conducted to analyze the effect of different hyperparameter configurations on the model's performance. The baseline YOLOv8 model recorded a box loss of 1.34, class loss of 1.87, and DFL loss of 1.47. Reducing the learning rate to 0.005 significantly improved performance, resulting in lower losses (box loss = 1.10, class loss = 1.45, and DFL loss = 1.22), indicating that a smaller learning rate enhances stable convergence. Conversely, increasing the learning rate to 0.02 led to higher losses compared to the reduced learning rate setting, highlighting the sensitivity of the model to overly aggressive learning rates. Similarly, increasing the batch size also had a positive effect on performance. With a batch size of 32, the losses decreased compared to the baseline, and further improvements were observed at a batch size of 64, where the box loss,

class loss, and DFL loss were minimized to 1.05, 1.40, and 1.15, respectively. These findings demonstrate that proper tuning of learning rate and batch size can substantially enhance the model's training stability and reduce overall loss.

In this study, YOLOv8's anchor-free detection mechanism was adopted, which allows the model to directly predict bounding box coordinates without relying on manually predefined anchor shapes. This approach provides greater flexibility in identifying irregularly shaped features in deforestation scenarios, such as fragmented tree stumps, equipment outlines, or clustered trunks. To further refine detection, the LangChain agent framework enabled dynamic adjustments of bounding box thresholds through reinforcement feedback, reducing misclassification and improving localization accuracy.

For performance evaluation, the framework was tested on both high-performance GPU and resource-constrained edge environments. On an NVIDIA RTX 3090 GPU, YOLOv8 achieved an average frame rate of ~45 FPS, ensuring real-time detection capability for continuous satellite or UAV image feeds. On embedded edge devices, frame rates averaged 15–20 FPS, which remained sufficient for real-time environmental monitoring and alert generation. These results highlight the framework's adaptability for both laboratory-scale experiments and field deployment.

In addition to detection accuracy, the framework demonstrated real-time applicability with an average frame rate of ~45 FPS on GPU and 15–20 FPS on edge devices, confirming its suitability for UAV- and satellite-based continuous forest monitoring.

Discussion

The LangChain Agent-enabled YOLOv8 model for anomaly identification demonstrates significant potential in real-time deforestation monitoring by combining YOLOv8's rapid object detection capabilities with LangChain's autonomous decision-making and reasoning framework. The model was evaluated using a diverse dataset containing visual indicators of deforestation, such as tree stumps, trunks, logging machinery, and unauthorized human presence. This integrated approach enhances real-time detection across large geographical areas while enabling adaptive decision-making in dynamic forest environments.

Key training metrics highlighted substantial improvements in localization and classification performance, with train/box_loss decreasing from ~1.3 to 0.5, train/cls_loss from 1.8 to 0.6, and train/dfl_loss from 1.5 to 1.0. However, validation losses exhibited a slight increase during later epochs, indicating potential overfitting—a challenge also reported in other object detection studies using small and imbalanced datasets. This aligns with findings in recent remote sensing research, where small object detection remains difficult due to limited training samples and background noise interference.

Comparative analysis with state-of-the-art models in AI-based deforestation monitoring shows that while our model achieved competitive detection of large objects (e.g., tree regions with recall up to 0.24), its mean average precision (mAP@0.5 = 0.071) and recall for smaller objects like stumps and equipment remain lower than advanced multispectral and transformer-based detection models. These results underscore the need for more robust feature extraction and balanced training data to mitigate false positives and improve detection of subtle deforestation indicators.

The confusion matrix analysis revealed frequent misclassification of smaller objects, with TREE TRUNK and equipment often confused with background regions. Similar misclassification patterns have been noted in other YOLO-based environmental monitoring studies, where low inter-class separability leads to reduced detection performance for minor features. To address this, future research should incorporate data augmentation, multispectral or SAR imagery, and meta-learning techniques to improve generalization under complex forest conditions.

The integration of LangChain agents provided an advantage over baseline YOLO models by enabling dynamic confidence adjustment and iterative retraining based on environmental feedback. This adaptive decision-making mechanism aligns with recent research advocating agentic AI for environmental applications, where reinforcement learning and feedback loops enhance detection robustness under evolving conditions. However, the model's current limitations—such as overfitting risk, conservative confidence thresholds, and weak small-object detection—indicate that further optimization through hyperparameter tuning, synthetic data generation, and cross-domain transfer learning is essential.

By critically comparing our findings with high-impact studies and acknowledging both strengths and weaknesses, this revised discussion provides a more balanced perspective. While the LangChain–YOLOv8 framework offers promising adaptability and real-time decision-making, improving dataset diversity, integrating multimodal inputs, and applying advanced optimization strategies remain crucial for achieving state-of-the-art performance in deforestation monitoring. These enhancements could ultimately establish the framework as a scalable and practical tool for sustainable forest management and global conservation efforts.

A key challenge in deforestation detection is the suppression of false positives, particularly in cases where visually similar features such as stumps, tools, or background textures are misclassified. To address this, the YOLOv8–LangChain framework integrates multiple suppression mechanisms. At the model level, Non-Maximum Suppression (NMS) is applied to remove redundant overlapping bounding boxes, ensuring only the most confident detections are preserved. Beyond this, the LangChain agent introduces dynamic threshold adjustment, where low-confidence or contextually inconsistent detections are flagged for further verification before being finalized. Furthermore, a reinforcement learning–based feedback loop enables the system to iteratively reduce false positives by incorporating corrected outputs into incremental retraining. For optimization, the model employed the AdamW optimizer with decoupled weight decay, chosen for its stability and efficiency on the moderately sized dataset. While methods such as SVRG and SGD were not implemented due to computational constraints, they represent promising directions for future work to enhance robustness against false positives.

For training stability, the YOLOv8 model was optimized using the AdamW optimizer with decoupled weight decay, which provided efficient convergence on the moderately sized dataset. In addition to standard Non-Maximum Suppression (NMS) applied at the model level, the integrated LangChain agent enabled dynamic threshold adjustment to suppress low-confidence or anomalous detections. A reinforcement learning–based feedback loop further reduced false positives by re-labeling misclassified instances and incorporating them into incremental retraining. While advanced stochastic methods such as Stochastic Variance Reduced Gradient (SVRG) and Stochastic Gradient Descent (SGD) were not implemented due to computational limitations, we acknowledge their potential for future enhancements in false positive suppression and overall model robustness.

Scalability for real-time deployment

The scalability of the YOLOv8–LangChain framework is a critical factor for its applicability in operational monitoring. Experimental results demonstrated that YOLOv8 achieved an average of ~45 FPS on GPU, while maintaining 15–20 FPS on edge devices such as NVIDIA Jetson, confirming feasibility for real-time UAV and satellite deployment. The modular separation of detection and reasoning enables the LangChain agent to asynchronously manage confidence threshold adjustment, reinforcement feedback, and GIS integration without compromising inference speed. Moreover, the framework supports horizontal scaling, where multiple UAV or satellite streams can be processed in parallel by distributed LangChain agents, transmitting only verified anomalies to a central hub. This design ensures efficient communication and responsiveness. Looking forward, integration with cloud services and federated learning can extend the system's scalability to regional or global forest monitoring.

Conclusion and future scope

The LangChain Agent-enabled YOLOv8 model has significant potential to detect anomalies linked to deforestation in forest settings, therefore assisting environmental sustainability projects. This approach aggregates the robust object detection framework of the YOLOv8 model with the autonomous decision-making capacity of LangChain Agent to effectively identify deforestation indicators like tree stumps, trunks, equipment, and human presence. Strong background identification performance of the algorithm highlights its ability to efficiently control large amounts of forested terrain. The model's poor recognition of smaller, less distinctive items, however, emphasizes the need of greater development to provide consistent identification under different environmental conditions.

The training results of the model exhibited good trends in lowering significant loss measures when train/box_loss, train/cls_loss, and train/dfl_loss dropped steadily over numerous epochs. These results indicate how, under the supervision of the model, learning and improvement are feasible all during training. Still, the rising trend in validation loss metrics alludes to probable overfitting and implies that the model demands improved generalizing methods. Techniques for data augmentation, more diverse datasets, and enhanced feature extraction help to tackle these challenges thereby boosting the real-world performance of the model.

Although the method can effectively identify background regions, its worse accuracy and recall values hint to clear limits in recognizing small signs of deforestation. Separating complex environmental patterns becomes challenging when the model leans to combine objects like TREE STUMP and TREE TRUNK with background data. These results highlight the need of complex object recognition techniques to guarantee correct mapping of deforestation trends and improve detection reliability.

LangChain agent framework offers fascinating avenues to enhance model adaption. The model will be able to adapt to evolving deforestation patterns by means of reinforcement learning approaches and continuous threshold adjustment in detection. Real-time feedback loops will also allow the system to switch detection techniques, hence reducing false positives and increasing accuracy. Such advances would provide useful information for conservation initiatives and assist the model to allow more effective monitoring of sensitive forest regions.

Though additional work is needed to increase its efficiency, the LangChain Agent-enabled YOLOV8 model demonstrates a significant development in the detection of deforestation. Improved performance results from improved generalizing capacity of the model, feature extraction techniques, and environmental change adaption capacity. These improvements will enable the model to become a major tool for monitoring deforestation activities, thus supporting environmental efforts, and so promoting sustainable development by methods of which it helps.

An important limitation of the present study is the lack of evaluation under low-light or nighttime conditions, since the dataset primarily consisted of daylight UAV and satellite imagery. Although data augmentation techniques such as brightness reduction and contrast variation were applied to simulate illumination changes, these cannot fully replicate nighttime scenarios. Future work will therefore focus on incorporating infrared (IR), thermal, and synthetic aperture radar (SAR) imagery to improve detection reliability in low-visibility environments. Integrating such multimodal data sources will enhance the robustness of the YOLOv8–LangChain framework, enabling continuous deforestation monitoring even during nighttime operations.

Future scope

Scalability, adaptability, and accuracy improvement of the LangChain Agent-enabled YOLOv8 model will enable it to have future scope in anomaly identification in deforestation monitoring. The ability of the model to detect smaller and less distinct objects including tree stumps, tools, and human presence determines the degree of comprehensive deforestation monitoring. Future research might focus on adopting innovative feature extraction methods like attention mechanisms or transformer-based architectures in order to increase the detection accuracy of the model. The model will be further able to identify patterns of deforestation under hard climatic conditions, like cloud cover and deep forest canopies, by using multi-modal data sources including LiDAR, multispectral, and synthetic aperture radar (SAR) pictures.

Another crucial issue for further studies is the investigation of adaptive learning methods allowing the model to evolve depending on changing deforestation trends. Combining reinforcement learning models will enable LangChain Agent to dynamically adjust detection thresholds and raise performance in demanding forest environments, hence enhancing its capacity for making decisions. Furthermore, using the model on edge devices such as satellites and drones allows real-time low latency monitoring of deforestation. This would help environmentalists and environmental authorities to respond fast to cases of deforestation, hence improving intervention strategies. By means of strengthening these skills, the LangChain Agent-enabled YOLOV8 model will become an indispensable tool in achieving long-term environmental sustainability and mitigating the effects of deforestation.

Data availability

The dataset will be available with the corresponding author based on individual requests.

Received: 8 June 2025; Accepted: 8 October 2025
Published online: 14 November 2025

References

1. Adjei, V. & Antwi, M. Land use change detection using the intensity analysis at the Bosomtwe District, Ashanti Region, Ghana. *Smart Agricultural Technol.* **5**, 100290. https://doi.org/10.1016/j.atech.2023.100290 (October 2023).
2. Akbari, V. & Solberg, S. Clear-Cut Detection and Mapping Using Sentinel-1 Backscatter Coefficient and Short-Term Interferometric Coherence Time Series. *IEEE Geosci. Remote Sens. Lett.* **19** : 1–5. https://doi.org/10.1109/LGRS.2020.3039875. (2022).
3. Almeida, J. et al. EdgeFireSmoke: A novel lightweight CNN model for Real-Time video Fire–Smoke detection. *IEEE Trans. Industr. Inf.* **18**, 11. https://doi.org/10.1109/TII.2021.3138752 (November 2022).
4. Almeida-Filho, R., Rosenqvist, A., Shimabukuro, Y. E. & dosSantos, J. R. Evaluation and perspectives of using multitemporal L-Band SAR data to monitor deforestation in the Brazilian AmazÔnia. *IEEE Geosci. Remote Sens. Lett.* **2** (4), 409–412. https://doi.org/10.1109/LGRS.2005.856679 (2005).
5. Alshehri, M., Ouadou, A., Grant, J. & Scott Deep Transformer-Based network deforestation detection in the Brazilian Amazon using Sentinel-2 imagery. *IEEE Geosci. Remote Sens. Lett.* **21**, 1–5. https://doi.org/10.1109/LGRS.2024.3355104 (2024).
6. Amani, M. et al. Wetland change analysis in Alberta, Canada using four decades of Landsat imagery. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **14**, 10314–10335. https://doi.org/10.1109/JSTARS.2021.3110460 (2021).
7. Amitrano, D., Guida, R. & Iervolino, P. Semantic Unsupervised Change Detection of Natural Land Cover With Multitemporal Object-Based Analysis on SAR Images. *IEEE Trans. Geosci. Remote Sens.* **59**, 7 : 5494–5514. https://doi.org/10.1109/TGRS.2020.3029841. (2021).
8. Borlido, I. et al. How to identify good superpixels for deforestation detection on tropical rainforests. *IEEE Geosci. Remote Sens. Lett.* **21**, 1–5. https://doi.org/10.1109/LGRS.2024.3454973 (2024).
9. Brooks, E. B. et al. On-the-Fly Massively Multitemporal Change Detection Using Statistical Quality Control Charts and Landsat Data. *IEEE Trans. Geosci. Remote Sens.* **52**, 6 : 3316–32. https://doi.org/10.1109/TGRS.2013.2272545. (2014).
10. Chatrabhuj & Meshram, K. Incremental learning model for sustainable agricultural land assessment using multimodal satellite data. *Int. J. Remote Sens.* **45**, 8622–8648. https://doi.org/10.1080/01431161.2024.2403628 (2024).
11. Chatrabhuj & Meshram, K. Design of an efficient satellite image analysis model for identification of sustainable construction areas via ground subsidence monitoring and infrastructure monitoring operations. *Indian Geotech. J.* **54** (3), 1136–1151 (2024).
12. Chatrabhuj & Meshram, K. Proposed design of an augmented deep learning model for Estimation of sustainable development goals via. *Satell. Image Anal. Environ. Dev. Sustain.* 1–27. https://doi.org/10.1007/s10668-023-04360-0 (2024).
13. Colditz, R. R., Ricardo, M., Llamas & Ressl, R. A. Detecting change areas in Mexico between 2005 and 2010 using 250 m MODIS images. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **7**, 8. https://doi.org/10.1109/JSTARS.2013.2280711 (2014).
14. Da Silva, E. & Freitas José Tasso Felix Guimarães, Gabriel Negreiros Salomão, Prafulla Kumar Sahoo, Douglas Batista Da Silva Ferreira, Mariana Maha Jana Costa Figueiredo, Silvio Junio Ramos. Multidecadal analysis of erosion susceptibility in a watershed heavily impacted by deforestation in southeastern Amazonia. *Int. J. Sediment. Res. Febr.* https://doi.org/10.1016/j.ijsrc.2025.01.010 (2025).
15. Dalal, S. et al. A deep learning framework with learning without forgetting for intelligent surveillance in IoT-enabled home environments in smart Cities. Recent advances in computer science and communications https://doi.org/10.2174/0126662558329951241024183922 (2024).
16. Dallaqua, F. B. J., Rojas, F. A. & Faria and Alvaro Luiz Fazenda. Building Data Sets for Rainforest Deforestation Detection Through a Citizen *Sci. Project IEEE Geoscience Remote Sens. Lett.* **19** : 1–5. https://doi.org/10.1109/LGRS.2020.3032098. (2022).
17. Das, P., Kumar, A., Sahu, D. V., Xavy & Meher, S. A deforestation detection network using deep Learning-Based semantic segmentation. *IEEE Sens. Lett.* **8** (1), 1–4. https://doi.org/10.1109/LSENS.2023.3340562 (2024).
18. Diniz, C. G., Souza, A. A. D. A. & Santos, D. C. Mirian Correa Dias, Nelton Cavalcante Da Luz, Douglas Rafael Vidal De Moraes, Janaina Sant Ana Maia. DETER-B: The New Amazon Near Real-Time Deforestation Detection System. *IEEE J. Selected Topics Appl. Earth Obs. Remote Sens.* 8, no. 7July : 3619–28 https://doi.org/10.1109/JSTARS.2015.2437075. (2015).
19. Du, Z. et al. Mapping Annual Global Forest Gain From 1983 to 2021 With Landsat Imagery. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **16** : 4195–4204. https://doi.org/10.1109/JSTARS.2023.3267796. (2023).
20. Eckert, S., Schmid, L., Messerli, P. & Zaehringer, J. G. Spatiotemporal assessment of deforestation and forest degradation indicates spillover effects from mining activities and related biodiversity offsets in Madagascar. *Remote Sens. Applications: Soc. Environ.* **36**, 101269. https://doi.org/10.1016/j.rsase.2024.101269 (November 2024).
21. Edeh, M. et al. A novel deep learning model for predicting marine pollution for sustainable ocean management. *PeerJ. Comput. Sci.* **10**, e2482 (2024).
22. Englhart, S., Keuck, V. & Siegert, F. Modeling aboveground biomass in tropical forests using Multi-Frequency SAR Data—A comparison of methods. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **5** (1), 298–306. https://doi.org/10.1109/JSTARS.2011.2176720 (2012).
23. Ferrari, F. & Ferreira, M. P. Cláudio Aparecido Almeida, and Raul Queiroz Feitosa. Fusing Sentinel-1 and Sentinel-2 images for deforestation detection in the Brazilian Amazon under diverse cloud conditions. *IEEE Geosci. Remote Sens. Lett.* **20**, 1–5. https://doi.org/10.1109/LGRS.2023.3242430 (2023).
24. Franke, J., Navratil, P., Keuck, V., Peterson, K. & Siegert, F. Monitoring fire and selective logging activities in tropical peat swamp forests. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **5**, 1811–1820. https://doi.org/10.1109/JSTARS.2012.2202638 (2012).
25. Gonçalves Sales, V. Modelling Non-Linear deforestation trends for an ecological tension zone in Brazil. *Sci. Remote Sens.* **7**, 100076. https://doi.org/10.1016/j.srs.2023.100076 (023).

26. Grings, F., Roitberg, E. & Barraza, V. EVI Time-Series breakpoint detection using convolutional networks for online deforestation monitoring in Chaco forest. *IEEE Trans. Geosci. Remote Sens.* **58** (2), 1303–1312. https://doi.org/10.1109/TGRS.2019.2945719 (2020).
27. Haq, B. et al. Tech-Driven forest conservation: combating deforestation with internet of Things, artificial Intelligence, and remote sensing. *IEEE Internet Things J.* **11**, 14. https://doi.org/10.1109/JIOT.2024.3378671 (2024).
28. Hermann, M., Saha, S. & Zhu, X. X. Filtering specialized change in a Few-Shot setting. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **16**, 1185–1196. https://doi.org/10.1109/JSTARS.2022.3231915 (2023).
29. Ji, M. et al. Deforestation detection from spaceborne Full-Waveform laser Altimetry, incorporating terrain effects: A case study in Porto Velho, Brazil. *Int. J. Appl. Earth Obs. Geoinf.* **129**, 103861. https://doi.org/10.1016/j.jag.2024.103861 (2024).
30. Kempeneers, P., Sedano, F., Strobl, P. & McInerney, D. O. Increasing robustness of postclassification change detection using time series of land cover maps. *IEEE Trans. Geosci. Remote Sens.* **50**, 9. https://doi.org/10.1109/TGRS.2011.2181854 (2012).
31. Lambin, E. F. & Linderman, M. Time Series of Remote Sensing Data for Land Change Science. *IEEE Trans. Geosci. Remote Sens.* **44**, 7 : 1926–28. https://doi.org/10.1109/TGRS.2006.872932. (2006).
32. Lilhore, U. et al. Intelligent waste sorting for sustainable environment: A hybrid deep learning and transfer learning model. *Gondwana Res.* **146**, 252–266. https://doi.org/10.1016/j.gr.2024.07.014 (2025).
33. Marcus, M. S., Hergoualc'h, K. & Eurídice, N. Honorio Coronado, and Víctor Hugo Gutiérrez-Vélez. Spatial distribution of degradation and deforestation of palm swamp peatlands and associated carbon emissions in the Peruvian Amazon. *J. Environ. Manage.* **351**, 119665. https://doi.org/10.1016/j.jenvman.2023.119665 (2024).
34. Mermoz, S. et al. Submonthly assessment of temperate forest Clear-Cuts in Mainland France. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **17**, 13743–13764. https://doi.org/10.1109/JSTARS.2024.3429012 (2024).
35. Meshram, K. Design of an iterative method for environmental-sustainable development. *Integrating Bioinspired Comput. Techniques Environ. Dev.* **51**, 101045. https://doi.org/10.1016/j.envdev.2024.101045 (2024).
36. Negri, R. G. et al. Unsupervised multitemporal triclass change detection. *IEEE Trans. Geosci. Remote Sens.* **62**, 1–17. https://doi.org/10.1109/TGRS.2024.3442156 (2024).
37. Olding, W. C., Jan, C., Olivier, Brian, P., Salmon & Kleynhans, W. A forecasting approach to online change detection in land cover time series. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **12**, 5. https://doi.org/10.1109/JSTARS.2019.2905594 (2019).
38. Parsch, C. et al. Forecasting deforestation and carbon loss across new Guinea using machine learning and cellular automata. *Sci. Total Environ.* **970**, 178864. https://doi.org/10.1016/j.scitotenv.2025.178864 (2025).
39. Radulescu, M., Dalal, S., Lilhore, U. K. & Sarita Saimiya. Optimizing mineral identification for sustainable resource extraction through hybrid deep learning enabled fintech model. *Resour. Policy.* **89**, 104692. https://doi.org/10.1016/j.resourpol.2024.104692 (2024).
40. Ramadan, M. N. A., Mohammed, A. H., Ali, S. Y., Khoo & Alkhedher, M. AI-Powered IoT and UAV systems for Real-Time detection and prevention of illegal logging. *Results Eng.* **24**, 103277. https://doi.org/10.1016/j.rineng.2024.103277 (2024).
41. Reiche, J. et al. Feature Level Fusion of Multi-Temporal ALOS PALSAR and Landsat Data for Mapping and Monitoring of Tropical Deforestation and Forest Degradation. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **6**, 5 : 2159–73. https://doi.org/10.1109/JSTARS.2013.2245101. (2013).
42. Schultz, M., Verbesselt, J., Avitabile, V., Souza, C. & Herold, M. Error sources in deforestation detection using BFAST monitor on Landsat time series across three tropical sites. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **9**, 8. https://doi.org/10.1109/JSTARS.2015.2477473 (2016).
43. Soja, M. J., Henrik, J., Persson, Lars, M. H. & Ulander Modeling and detection of deforestation and forest growth in multitemporal TanDEM-X data. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **11**, 10. https://doi.org/10.1109/JSTARS.2018.2851030 (2018).
44. Son, N. T., Chen, C. F., Chang, N. B. & Chen, C. R. Ly-Yu Chang, and Bui-Xuan Thanh. Mangrove Mapping and Change Detection in Ca Mau Peninsula, Vietnam, Using Landsat Data and Object-Based Image Analysis. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **8**, 2 : 503–510. https://doi.org/10.1109/JSTARS.2014.2360691. (2015).
45. Soto, P. J. et al. Domain-Adversarial neural networks for deforestation detection in tropical forests. *IEEE Geosci. Remote Sens. Lett.* **19**, 1–5. https://doi.org/10.1109/LGRS.2022.3163575 (2022).
46. Sugimoto, R., Natsuaki, R., Nakamura, R., Tsutsumi, C. & Yamaguchi, Y. Time series scattering power decomposition using ensemble average in Temporal–Spatial domains: application to forest disturbance detection. *IEEE Geosci. Remote Sens. Lett.* **21**, 1–5. https://doi.org/10.1109/LGRS.2023.3346378 (2024).
47. Tan, K., Jin, X., Plaza, A., Wang, X., Du, P. & Liang Xiao, and Automatic change detection in High-Resolution remote sensing images by using a multiple classifier system and Spectral–Spatial features. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **9**, 8. https://doi.org/10.1109/JSTARS.2016.2541678 (2016).
48. Coronel, T. Mapping deforestation using fractions indices and the Non-Seasonal PVts- β detection approach. *IEEE Geosci. Remote Sens. Lett.* **19**, 1–5. https://doi.org/10.1109/LGRS.2021.3137277 (2022).
49. Urugo, M., Makiso, M., Worku, Y. B. & Tola Ethiopian coffee: production Systems, geographical origin Traceability, and European union deforestation regulation directive compliance. *J. Agric. Food Res.* **19**, 101695. https://doi.org/10.1016/j.jafr.2025.101695 (2025).
50. Valderrama-Landeros, L., Humberto, M. L., Baret, F. & Time Series, C. Y. C. L. O. P. E. S. L. A. I. Espana-Boquera, andDeforestation in Michoacan, Mexico, From (2000–2006). IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 9, no. 12December : 5398–5405. https://doi.org/10.1109/JSTARS.2016.2597742 (2016).
51. Vega, P. J. & Soto Gilson Alexandre Ostwald Pedro Da Costa, Mabel Ximena Ortega Adarme, José David Bermúdez Castro, and Raul Queiroz Feitosa. Weakly supervised domain adversarial neural network for deforestation detection in tropical forests. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **16**, 10264–10278. https://doi.org/10.1109/JSTARS.2023.3327573 (2023).
52. Verma, H. et al. Development of LR-PCA based fusion approach to detect the changes in Mango fruit crop by using Landsat 8 OLI images. *IEEE Access.* **10**, 85764–85776. https://doi.org/10.1109/ACCESS.2022.3194000 (2022).
53. Walker, W. S., Stickler, C. M., Kellndorfer, J. M., Kirsch, K. M. & Nepstad, D. C. Large-Area Classification and Mapping of Forest and Land Cover in the Brazilian Amazon: A Comparative Analysis of ALOS/PALSAR and Landsat Data Sources. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **3**, 4 : 594–604. https://doi.org/10.1109/JSTARS.2010.2076398. (2010).
54. Wang, Z. et al. A Web-Based prototype system for deforestation detection on High-Resolution remote sensing imagery with deep learning. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **17**, 18593–18612. https://doi.org/10.1109/JSTARS.2024.3435372 (2024).
55. Watanabe, M., Koyama, C. N., Hayashi, M., Nagatani, I. & Shimada, M. Early-Stage deforestation detection in the tropics with L -Band SAR. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **11**, 6. https://doi.org/10.1109/JSTARS.2018.2810857 (2018).
56. Yang, B. et al. Sub-Annual Scale LandTrendr: Sub-Annual Scale Deforestation Detection Algorithm Using Multi-Source Time Series Data. *IEEE J. Sel. Top. Appl. Earth Observations Remote Sens.* **16** : 8563–8576. https://doi.org/10.1109/JSTARS.2023.3312812. (2023).
57. Zanotta, D., Capella, L., Bruzzone, F. & Bovolo An adaptive semisupervised approach to the detection of User-Defined recurrent changes in image time series. *IEEE Trans. Geosci. Remote Sens.* **53**, 7. https://doi.org/10.1109/TGRS.2014.2381645 (2015).

Author contributions

SK & SZN were responsible for Validation, Software, Data Curation, and Writing - Original Draft. RPC, MD & AM were responsible for Conceptualization, Writing - Original Draft. RB, AA & AH were responsible for Writing the original Draft and visualization. AA & AH were responsible for Writing - Review & Editing. SK, RB & RPC were responsible for Formal Analysis. AH & RB was responsible for Writing - Original Draft, Resources, and Supervision. The author(s) read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.D.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025